Masalah Transportasi (*Transportation Problem*)

Kuliah 12

Materi Bahasan

① Rumusan masalah transportasi

② Pemecahan masalah transportasi

① Rumusan Masalah Transportasi

Masalah Transportasi

- Masalah transportasi umumnya berkaitan dengan masalah pendistribusian suatu produk dari beberapa sumber ke sejumlah tujuan dengan biaya yang minimum.

Rumusan Pemrograman Linier (1)

- Terdapat *m* sumber (misal: gudang) dimana produk disimpan.
- Terdapat *n* tujuan (misal: pasar) dimana produk dibutuhkan.
- Ketersediaan pasokan dari sumber : a_i ($i = 1, 2, \ldots, m$)
- Permintaan dari tujuan : b_j ($j = 1, 2, \ldots, n$)
- Biaya pengiriman dari sumber *i* ke tujuan *j* : c_{ij} ($i = 1, 2, \ldots, m$; $j = 1, 2, \ldots, n$). Jika suatu sumber *i* tidak dapat memasok suatu tujuan *j*, maka $c_{ij} = M$ (*M* bilangan positif yang sangat besar)
- Permasalahannya adalah menentukan jumlah produk yang dikirim dari sumber *i* ke tujuan *j* (dinyatakan dengan x_{ij}) yang meminimumkan biaya transportasi (pengiriman) total.

Rumusan Pemrograman Linier (2)

Minimasi $$Z = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij}$$

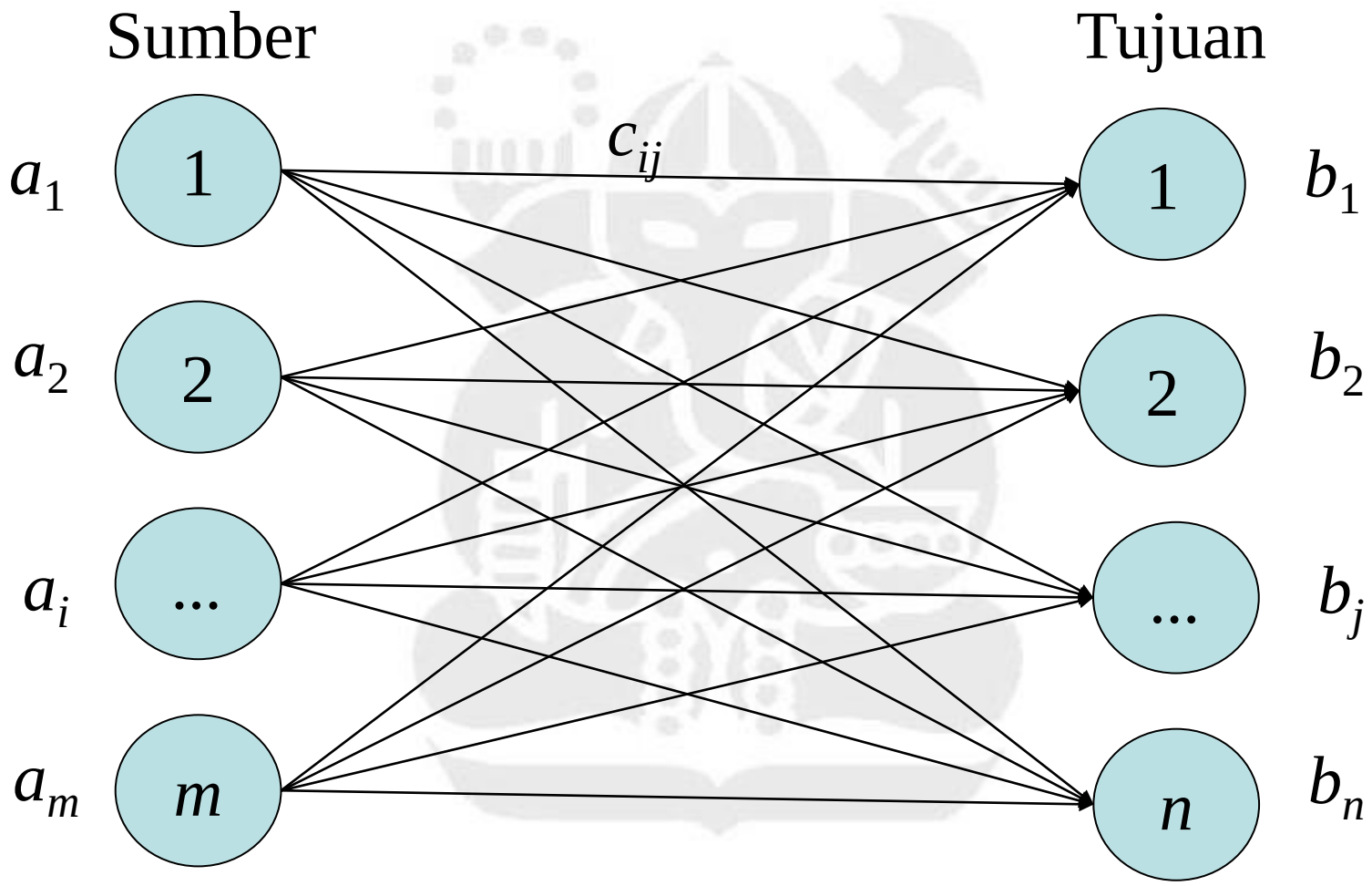
dengan pembatas-pembatas:

$$\sum_{j=1}^{n} x_{ij} \le a_i \qquad i = 1, 2, \ldots, m$$

$$\sum_{i=1}^{m} x_{ij} \ge b_j \qquad j = 1, 2, \ldots, n$$

$$x_{ij} \ge 0 \qquad \forall i, j$$

Masalah Transportasi dalam Bentuk Jaringan



Masalah Transportasi Standar/Seimbang (*Standar/Balanced Transportation Problem*)

Minimasi $$Z = \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij} x_{ij}$$

$$\sum_{j=1}^{m} a_i = \sum_{i=1}^{n} b_j$$

dengan pembatas-pembatas:

$$\sum_{j=1}^{n} x_{ij} = a_i \qquad i = 1, 2, \ldots, m$$

$$\sum_{i=1}^{m} x_{ij} = b_j \qquad j = 1, 2, \ldots, n$$

$$x_{ij} \geq 0 \qquad \forall i, j$$

Masalah Transportasi Tak Seimbang (1)

$$\sum_{j=1}^{m} a_i > \sum_{i=1}^{n} b_j$$

Minimasi $$Z = \sum_{i=1}^{m}\sum_{j=1}^{n+1} c_{ij} x_{ij}$$

dengan pembatas-pembatas:

$$\sum_{j=1}^{n+1} x_{ij} = a_i \qquad i = 1, 2, \ldots, m$$

$$\sum_{i=1}^{m} x_{ij} = b_j \qquad j = 1, 2, \ldots, n, n+1$$

$$x_{ij} \geq 0 \qquad \forall i, j$$

$j = n+1$ adalah tujuan fiktif dengan permintaan $b_{n+1} = \sum_{j=1}^{m} a_i - \sum_{i=1}^{n} b_j$

dan $c_{i,n+1} = 0,\ i = 1,2,...,m$

Masalah Transportasi Tak Seimbang (2)

$$\sum_{j=1}^{m} a_i < \sum_{i=1}^{n} b_j$$

Minimasi $$Z = \sum_{i=1}^{m+1} \sum_{j=1}^{n} c_{ij} x_{ij}$$

dengan pembatas-pembatas:

$$\sum_{j=1}^{n} x_{ij} = a_i \qquad i = 1, 2, \ldots, m+1$$

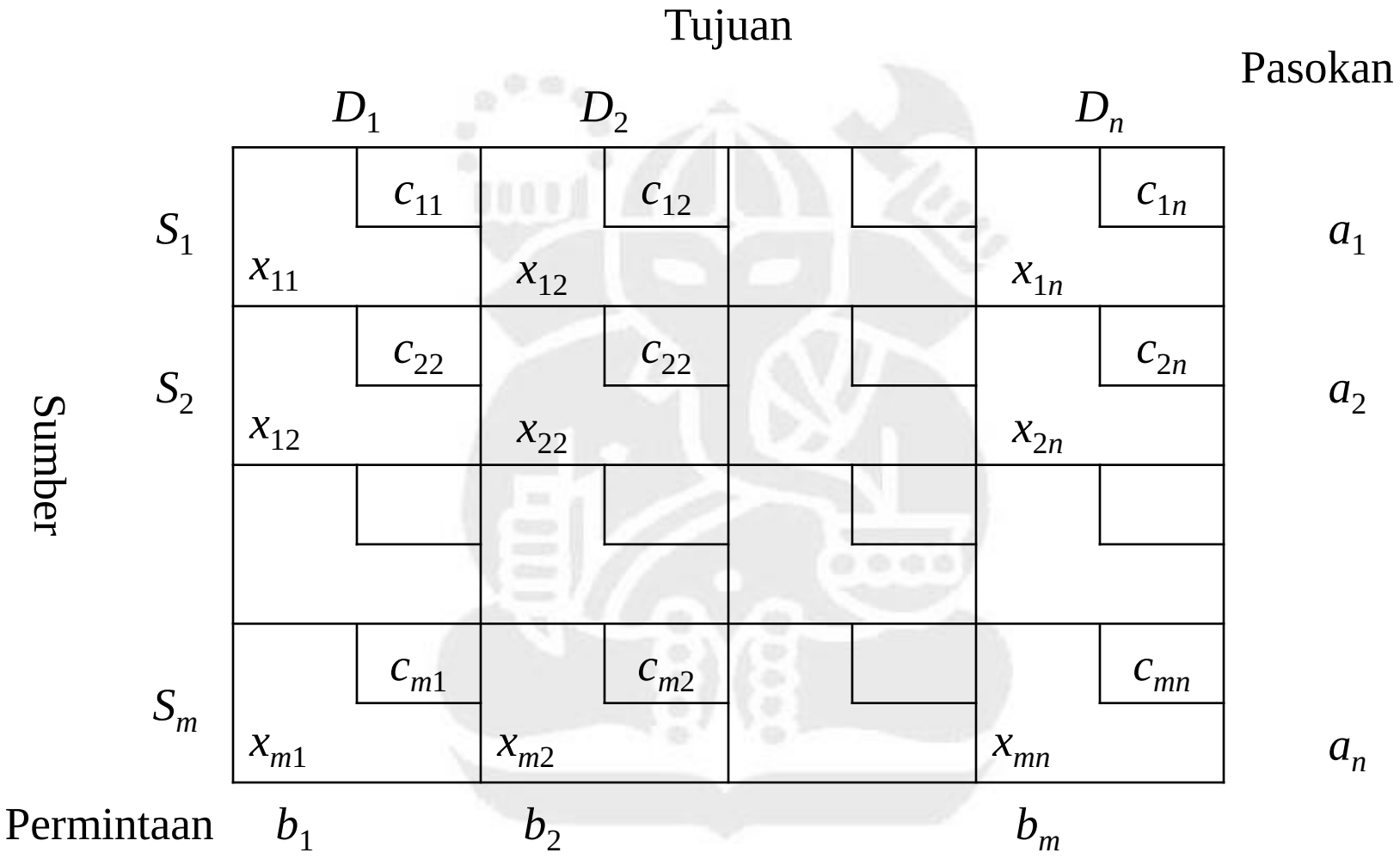
$$\sum_{i=1}^{m+1} x_{ij} = b_j \qquad j = 1, 2, \ldots, n$$

$$x_{ij} \geq 0 \qquad \forall i, j$$

$j = n+1$ adalah tujuan fiktif dengan permintaan $b_{n+1} = \sum_{i=1}^{n} b_j - \sum_{j=1}^{m} a_i$

dan $c_{m+1,\,j} = 0, \;\; j = 1,2,...,n$

Tabel Transportasi



② Pemecahan Masalah Transportasi

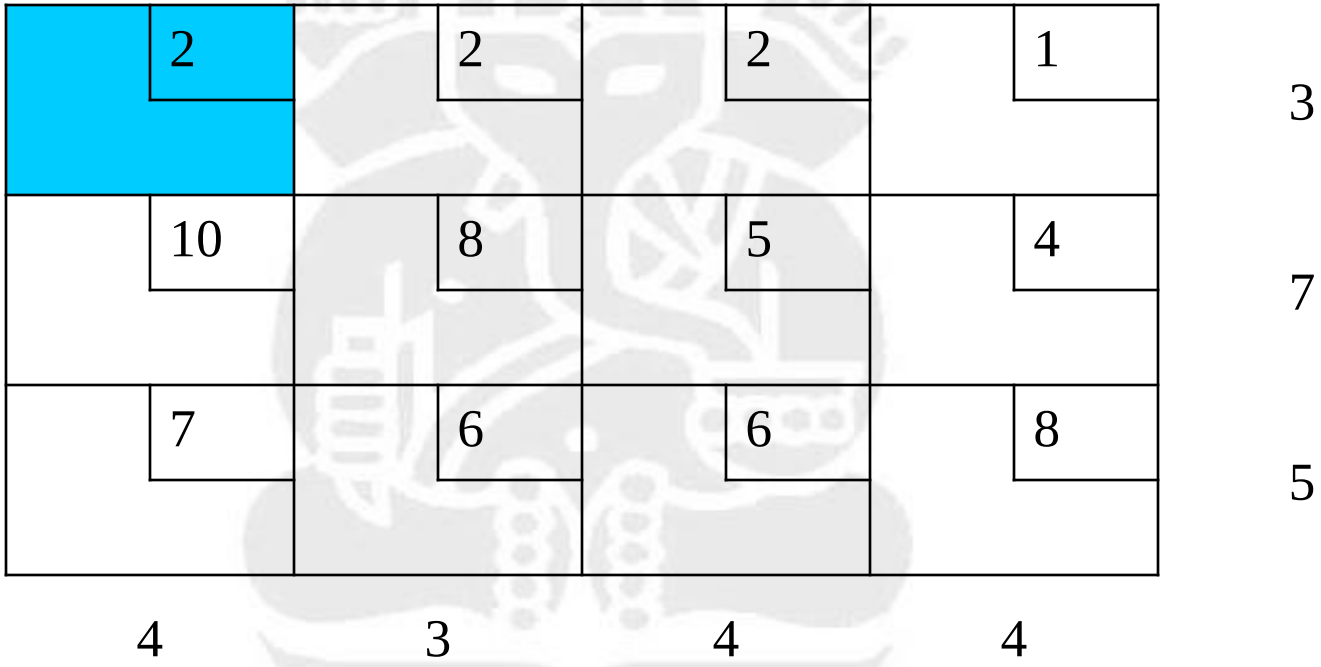
Algoritma Pemecahan Masalah Transportasi

- Langkah 0:
  - Perumusan masalah dalam masalah transportasi standar
- Langkah 1:
  - Penentuan solusi basis layak awal
- Langkah 2:
  - Pemeriksaan optimalitas. Jika solusi optimal maka berhenti.
  - Penentuan solusi basis yang baru dan ke langkah 2

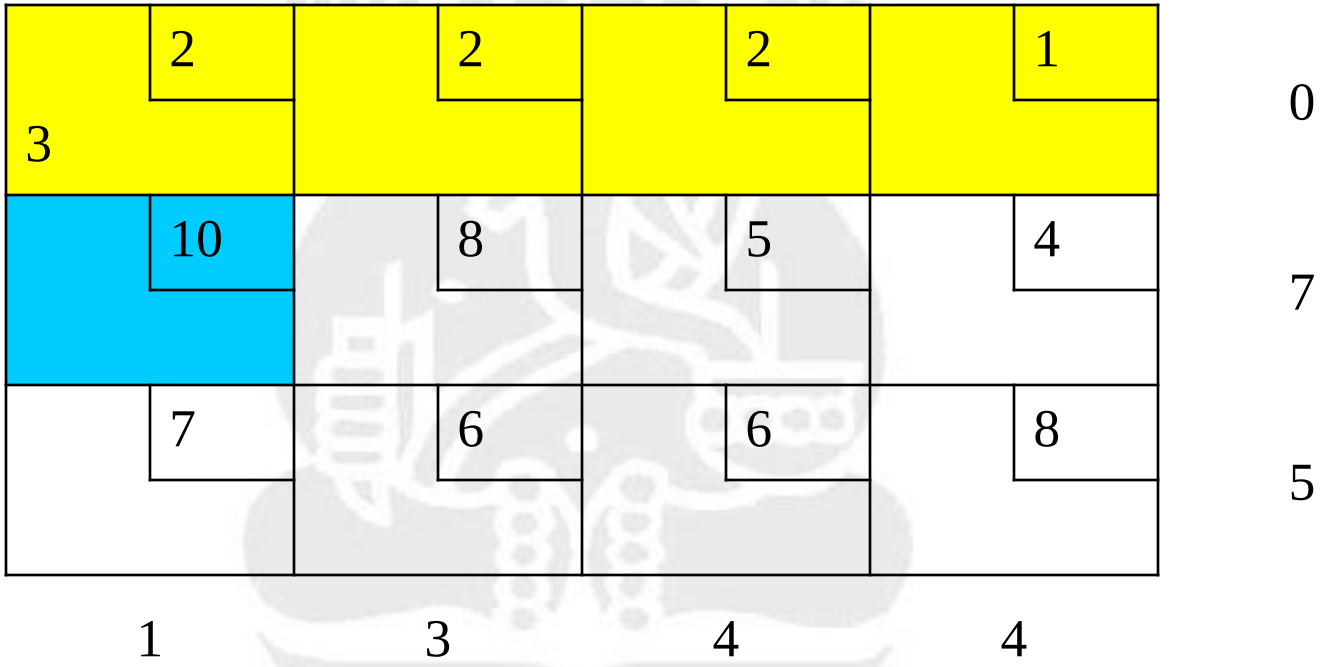
Metode Penentuan Solusi Basis Layak Awal

- *Northwest corner method*
- *Row minimum method*
- *Column minimum method*
- *Least cost method* (*Matrix minimum method*)
- *Vogel's approximation method* (VAM)
- *Russell's approximation method* (RAM)

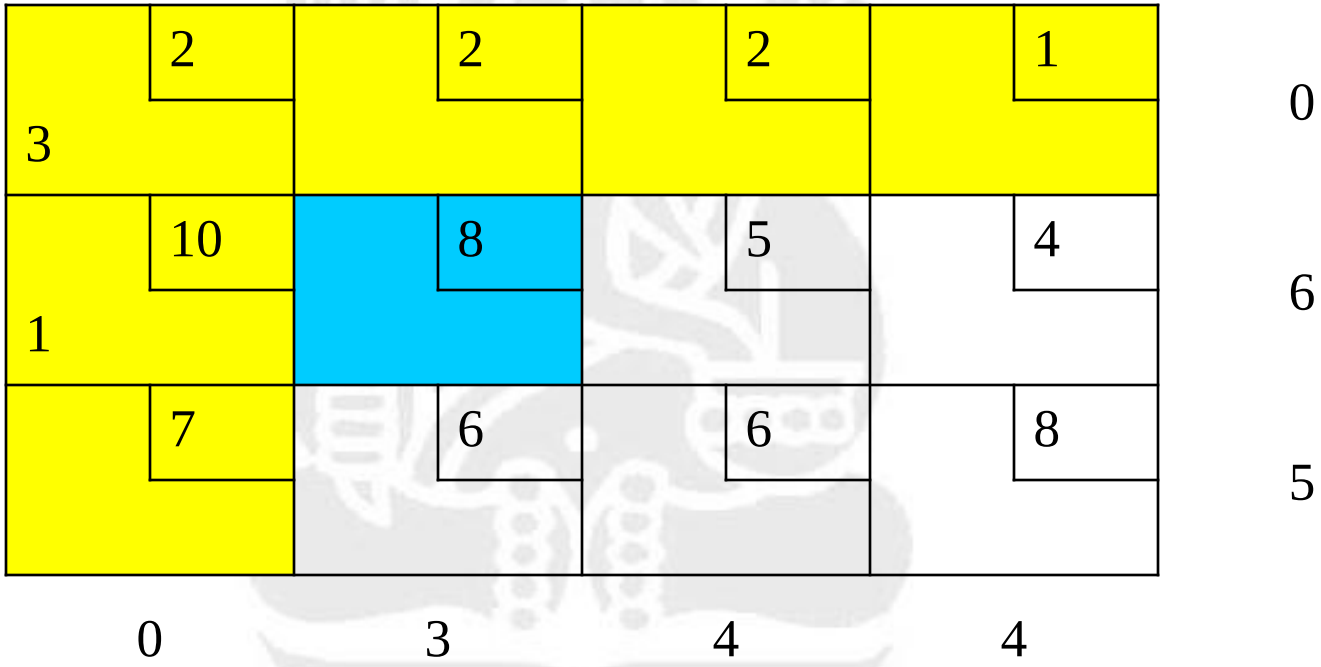
Northwest Corner Rule (1)



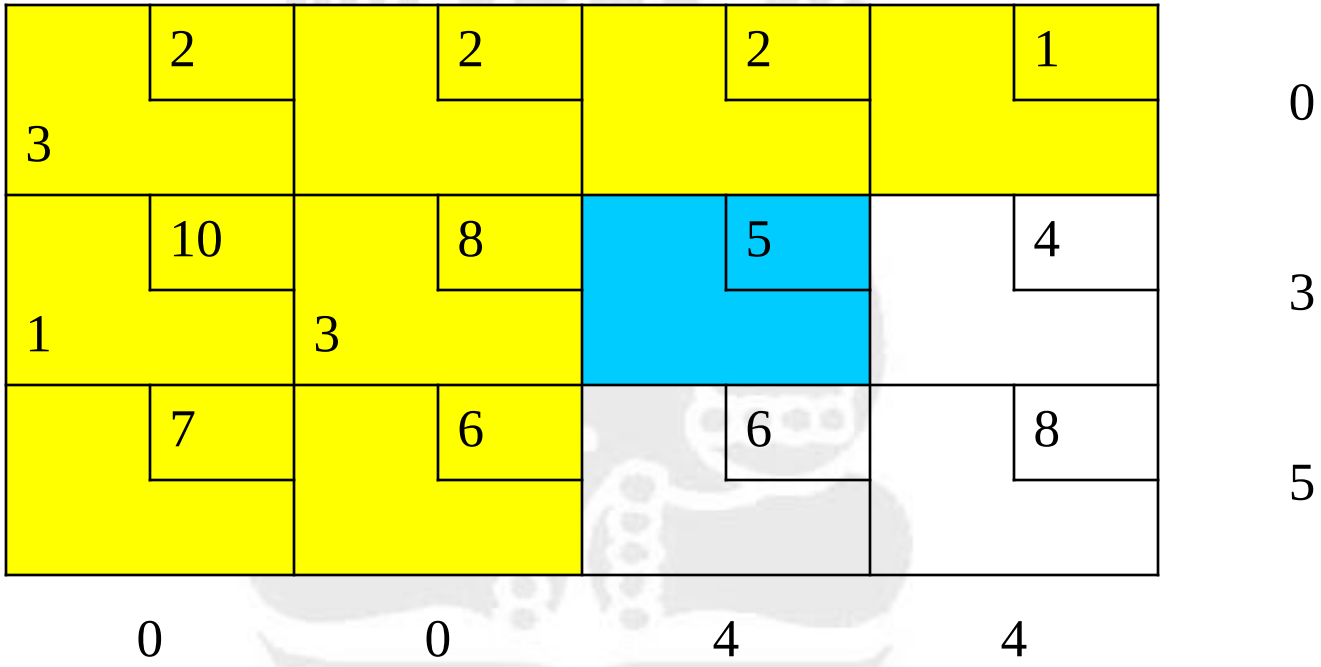
Northwest Corner Rule (2)



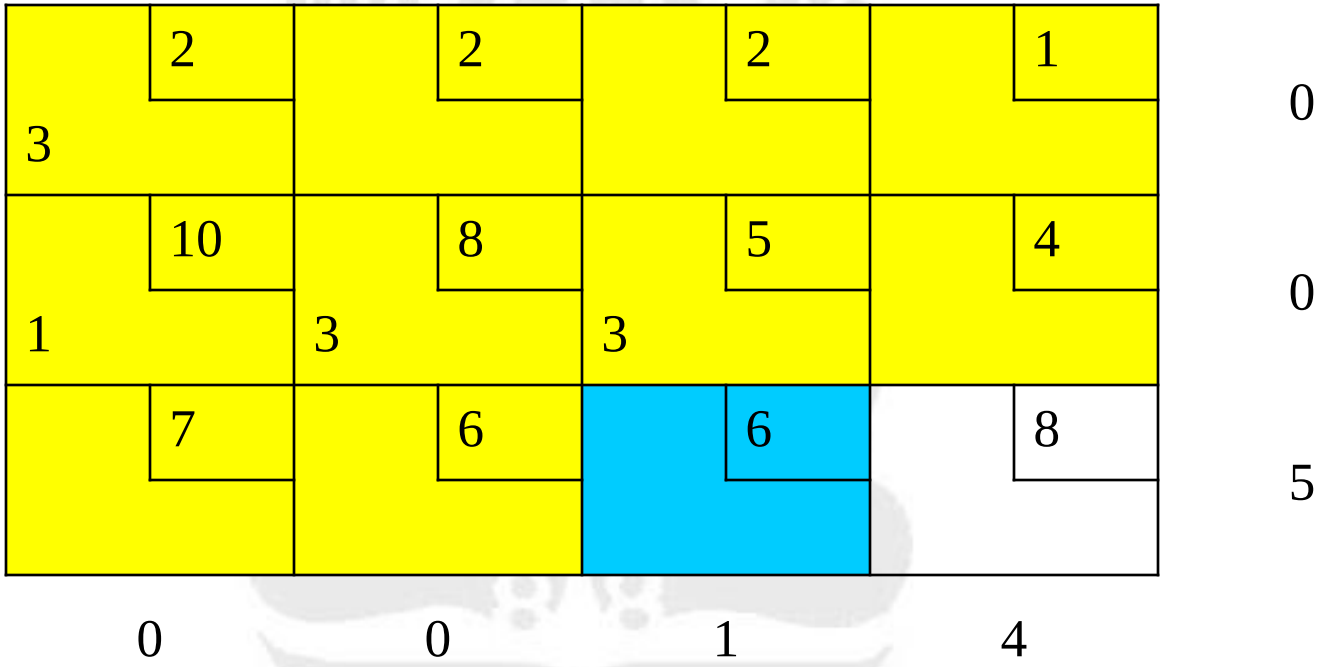
Northwest Corner Rule (3)



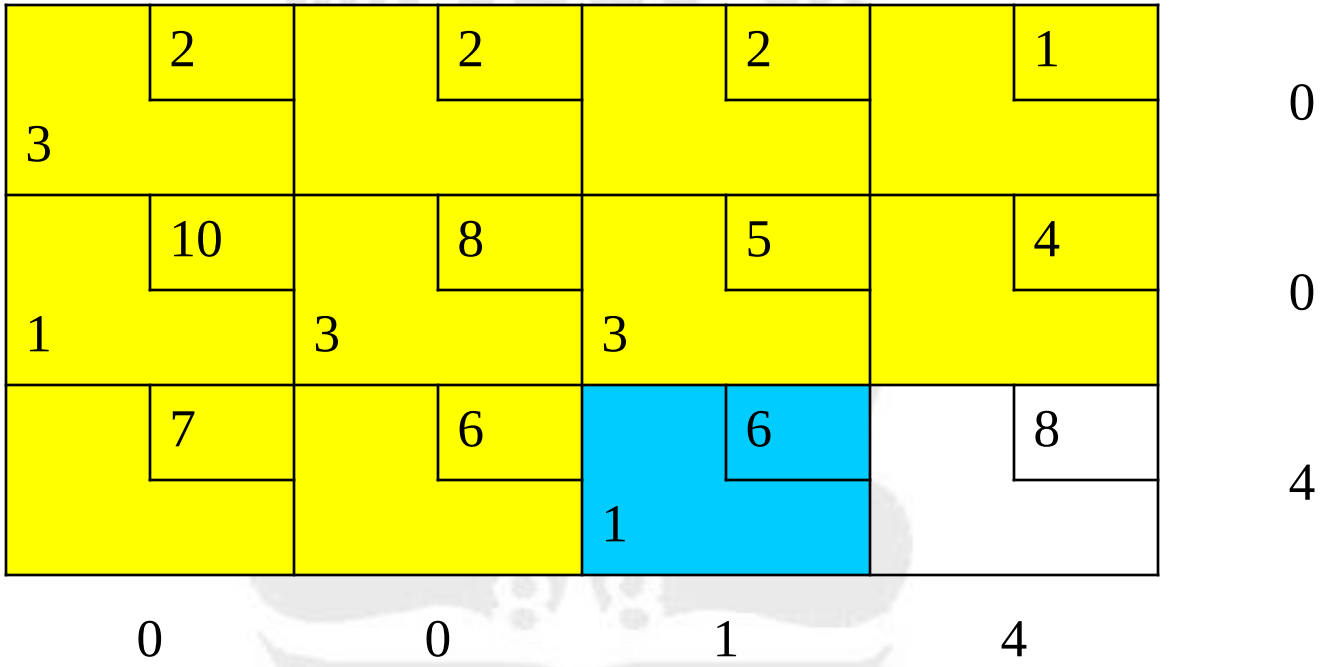
Northwest Corner Rule (4)



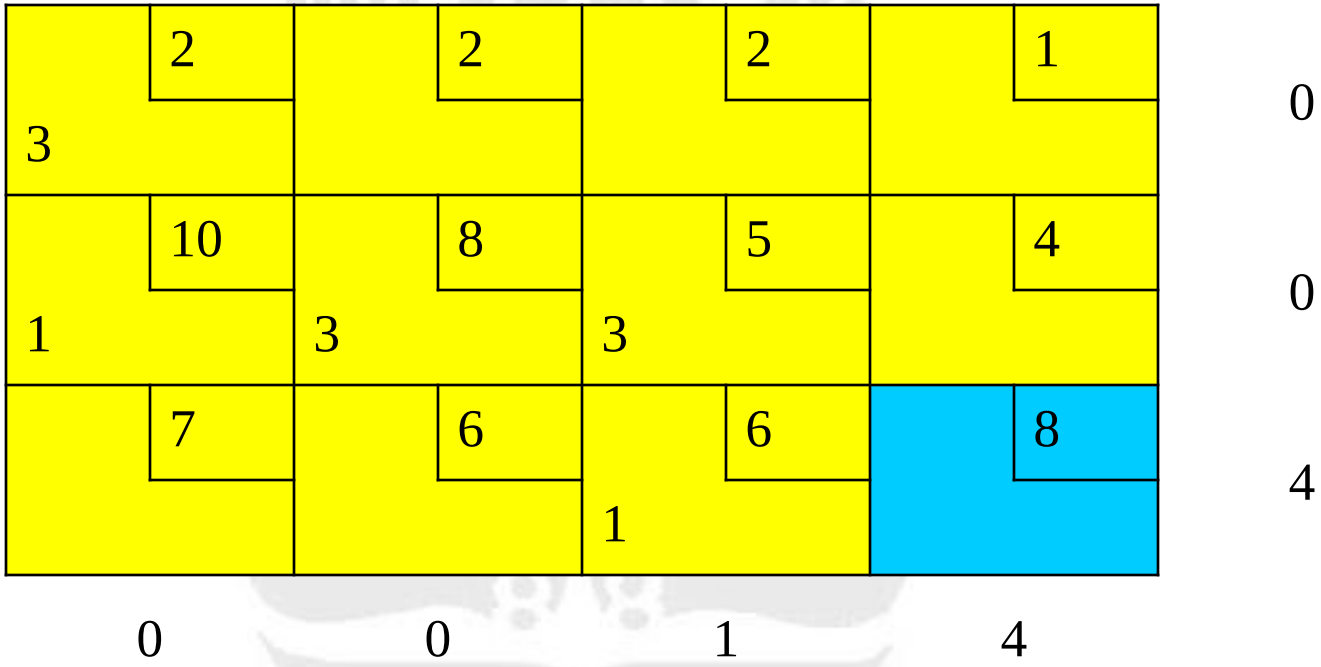
Northwest Corner Rule (5)



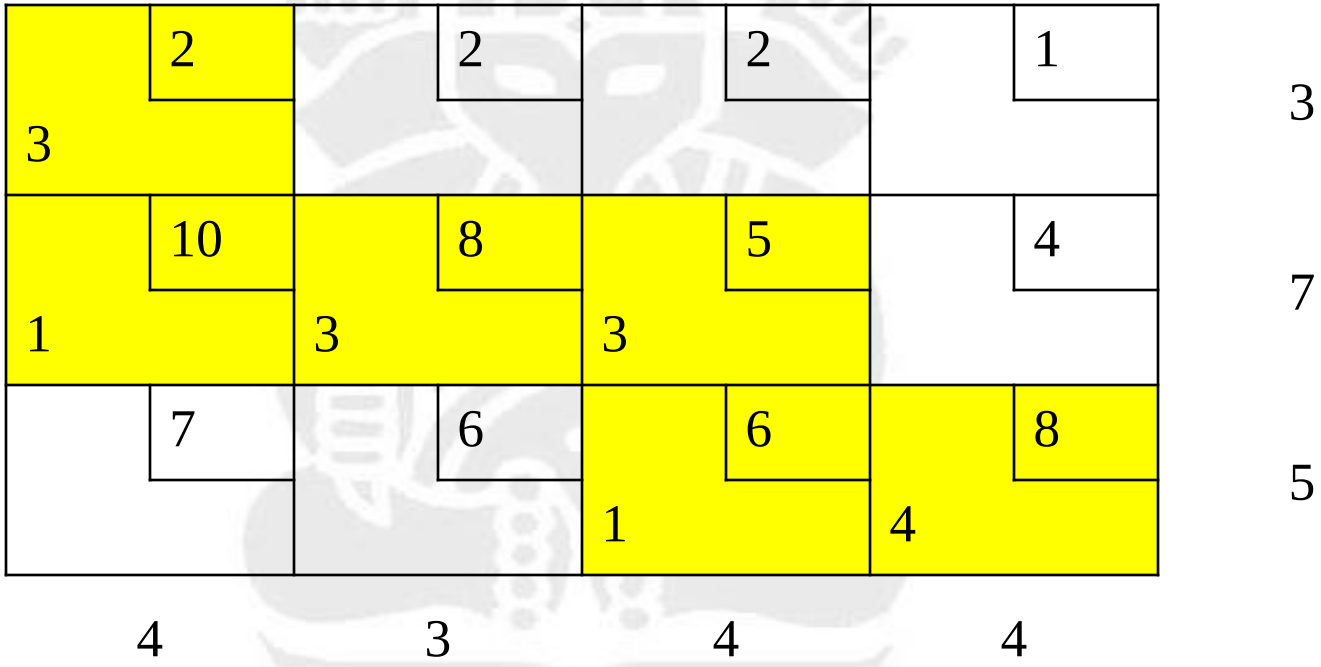
Northwest Corner Rule (5)



Northwest Corner Rule (6)

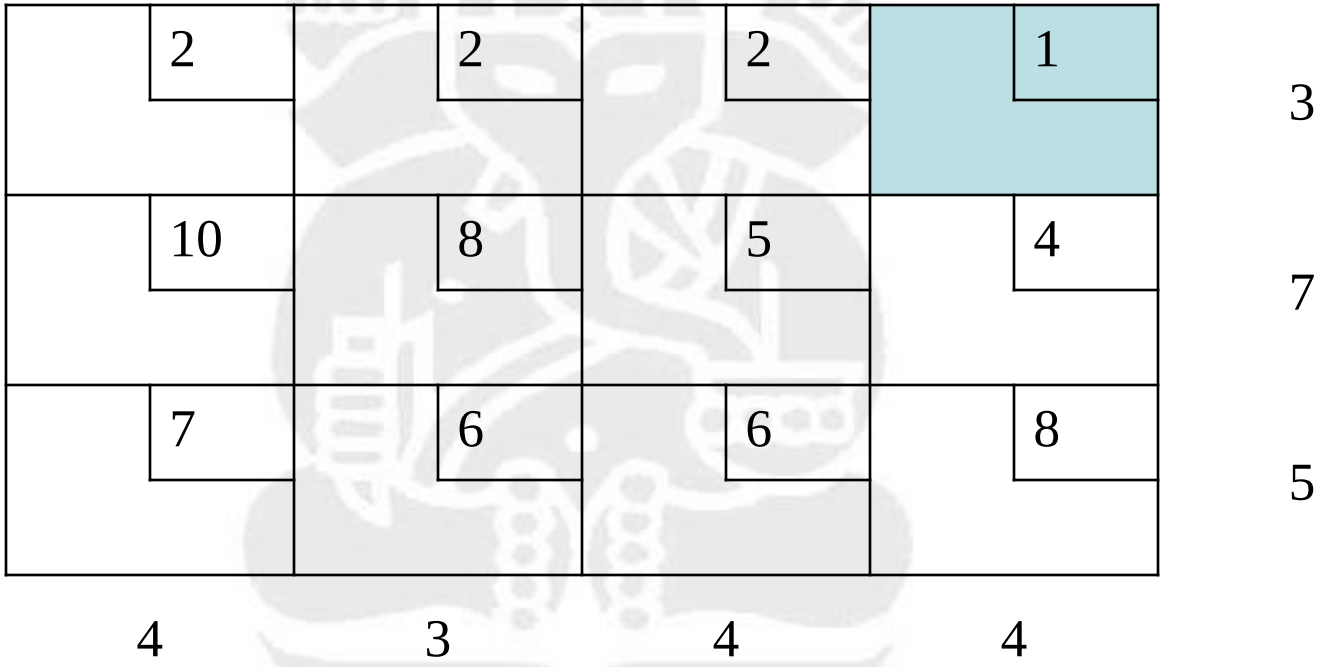


Northwest Corner Rule (6)
Solusi Basis Layak Awal

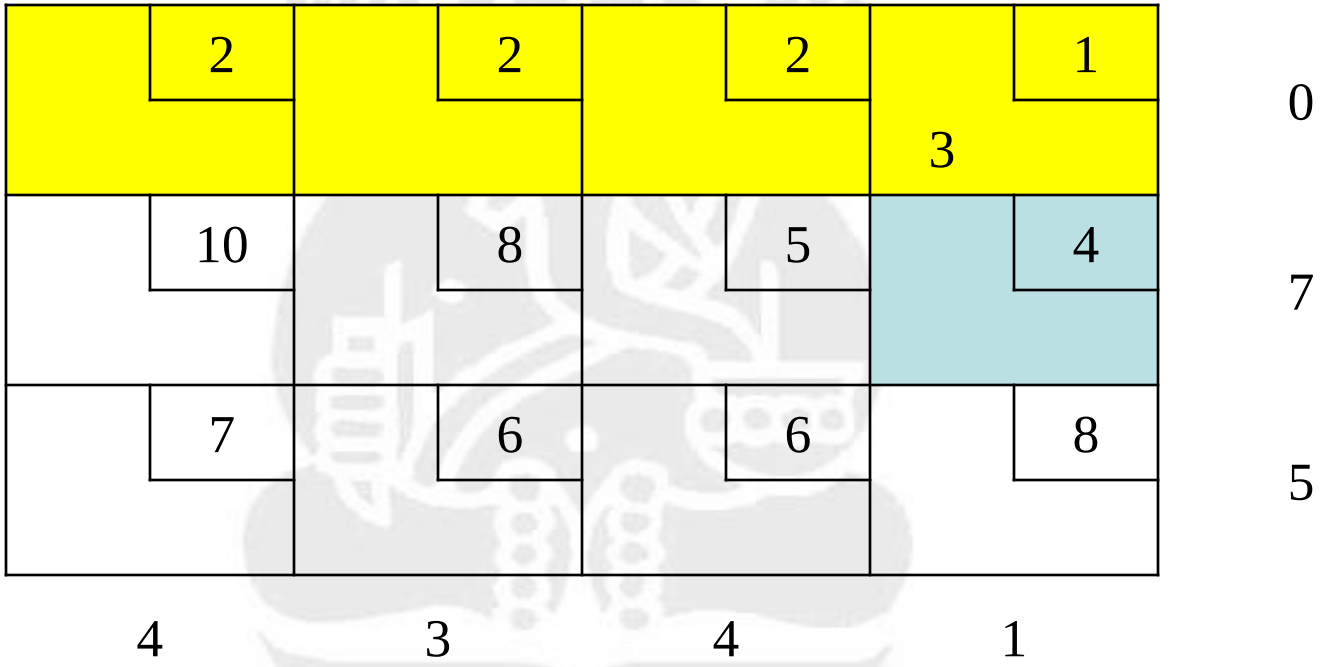


Biaya transportasi total $Z = 92$

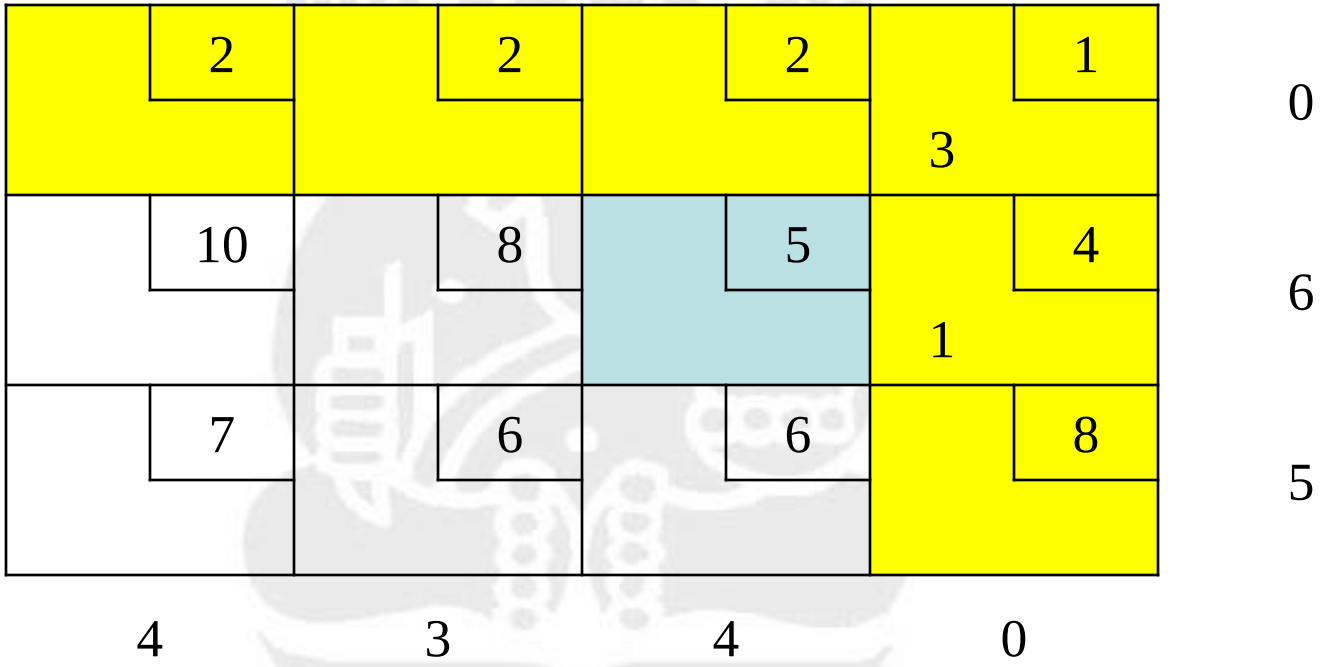
Least Cost Rule (1)



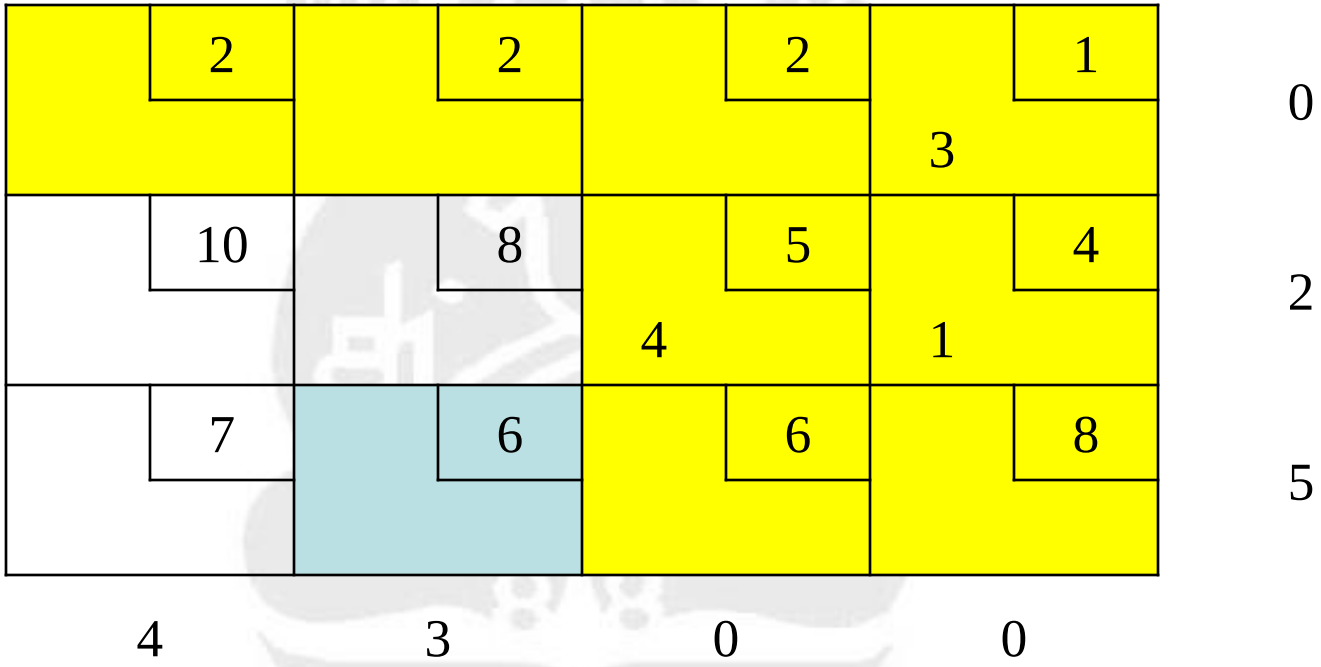
Least Cost Rule (2)



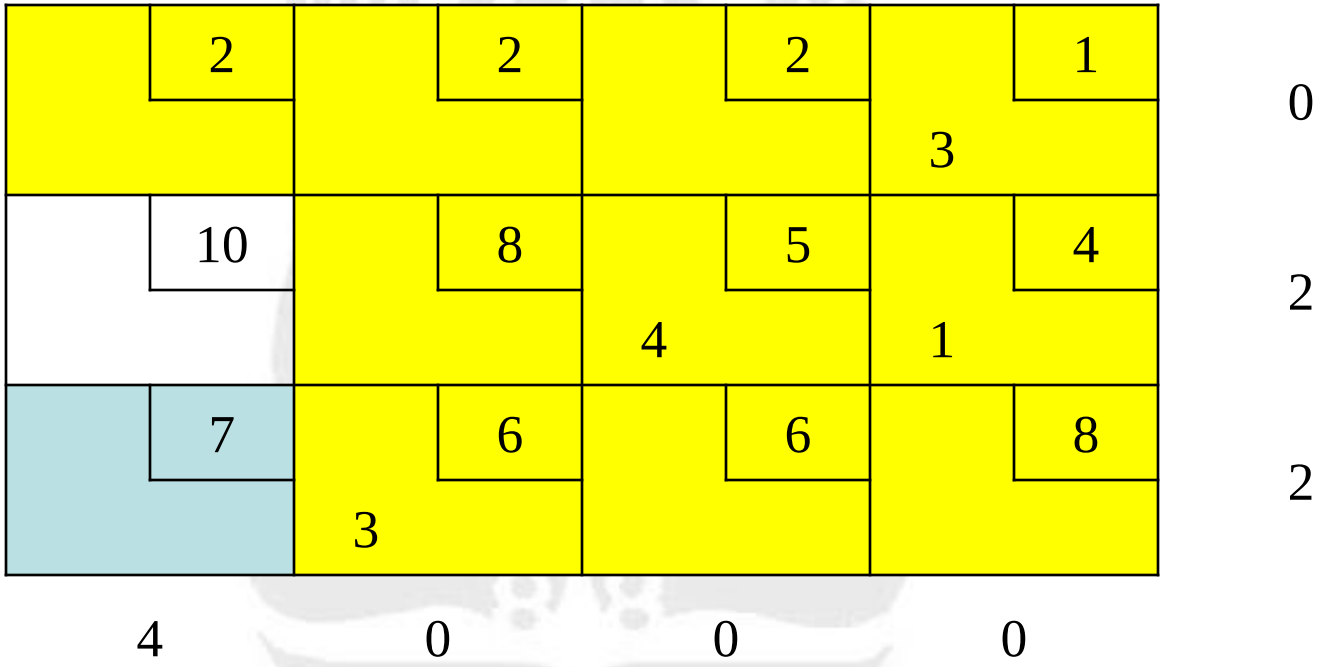
Least Cost Rule (3)



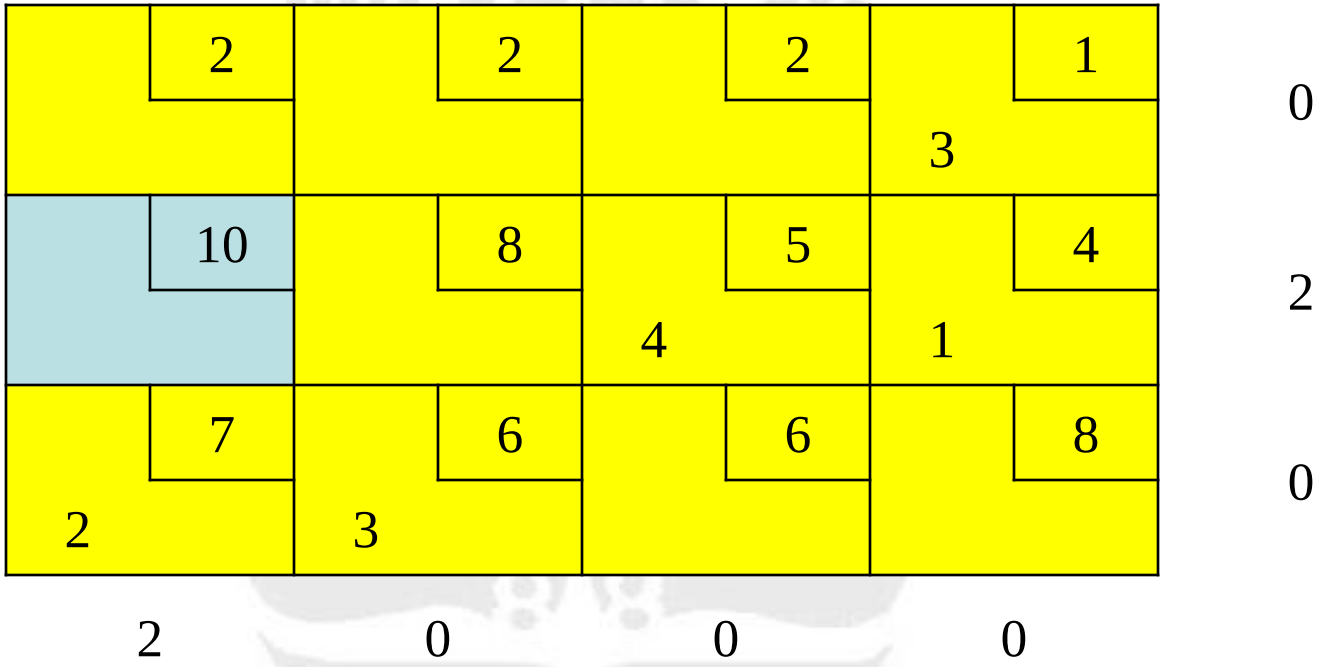
Least Cost Rule (4)



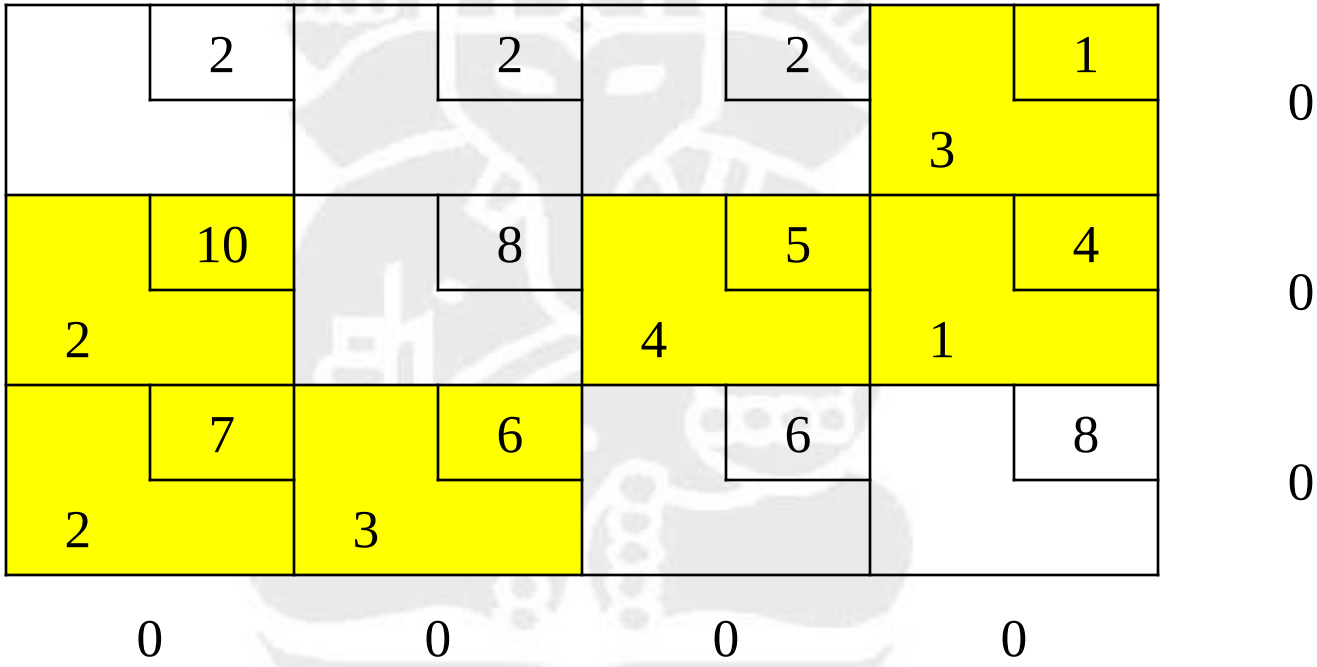
Least Cost Rule (5)



Least Cost Rule (6)

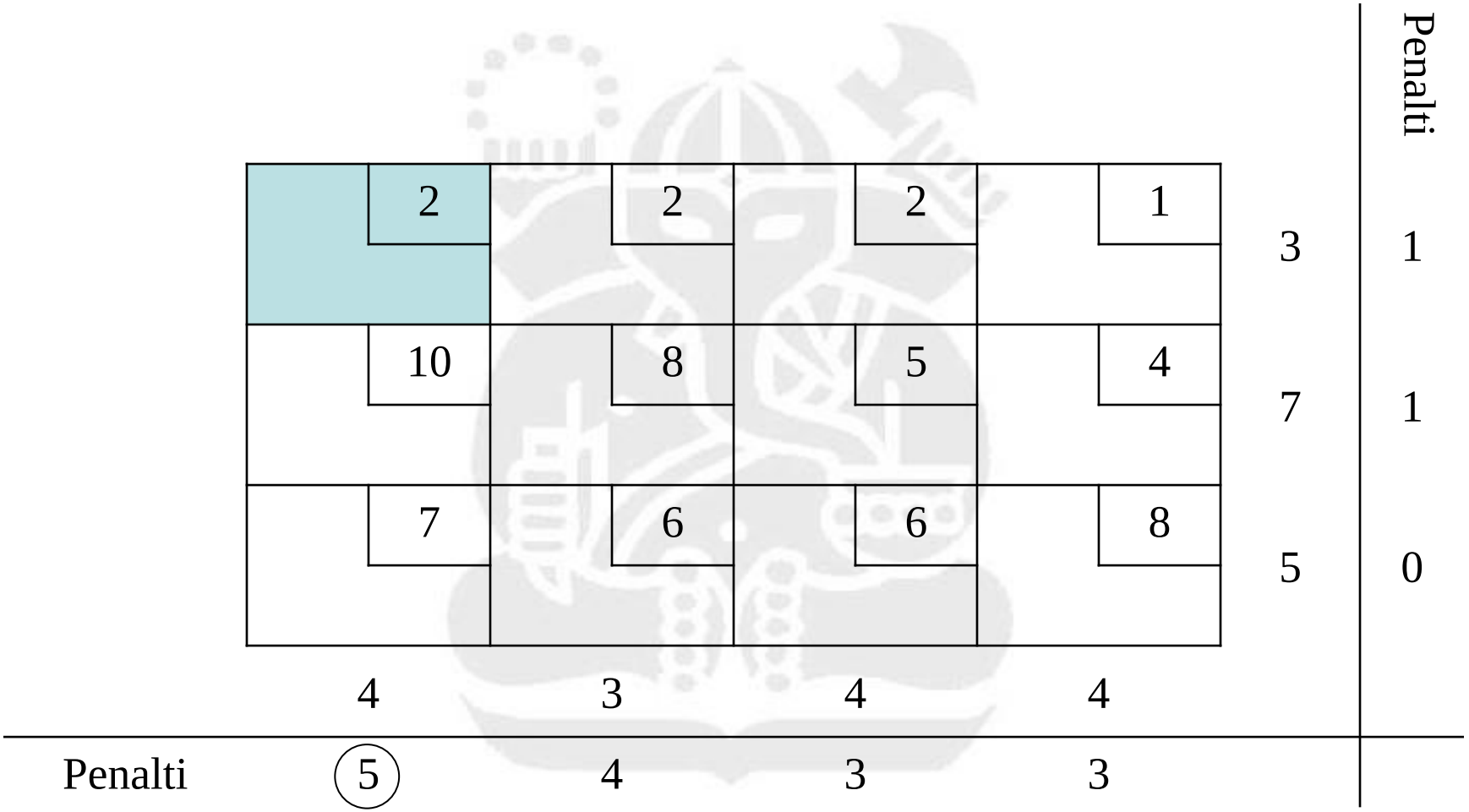


Least Cost Rule (7)
Solusi Basis Layak Awal

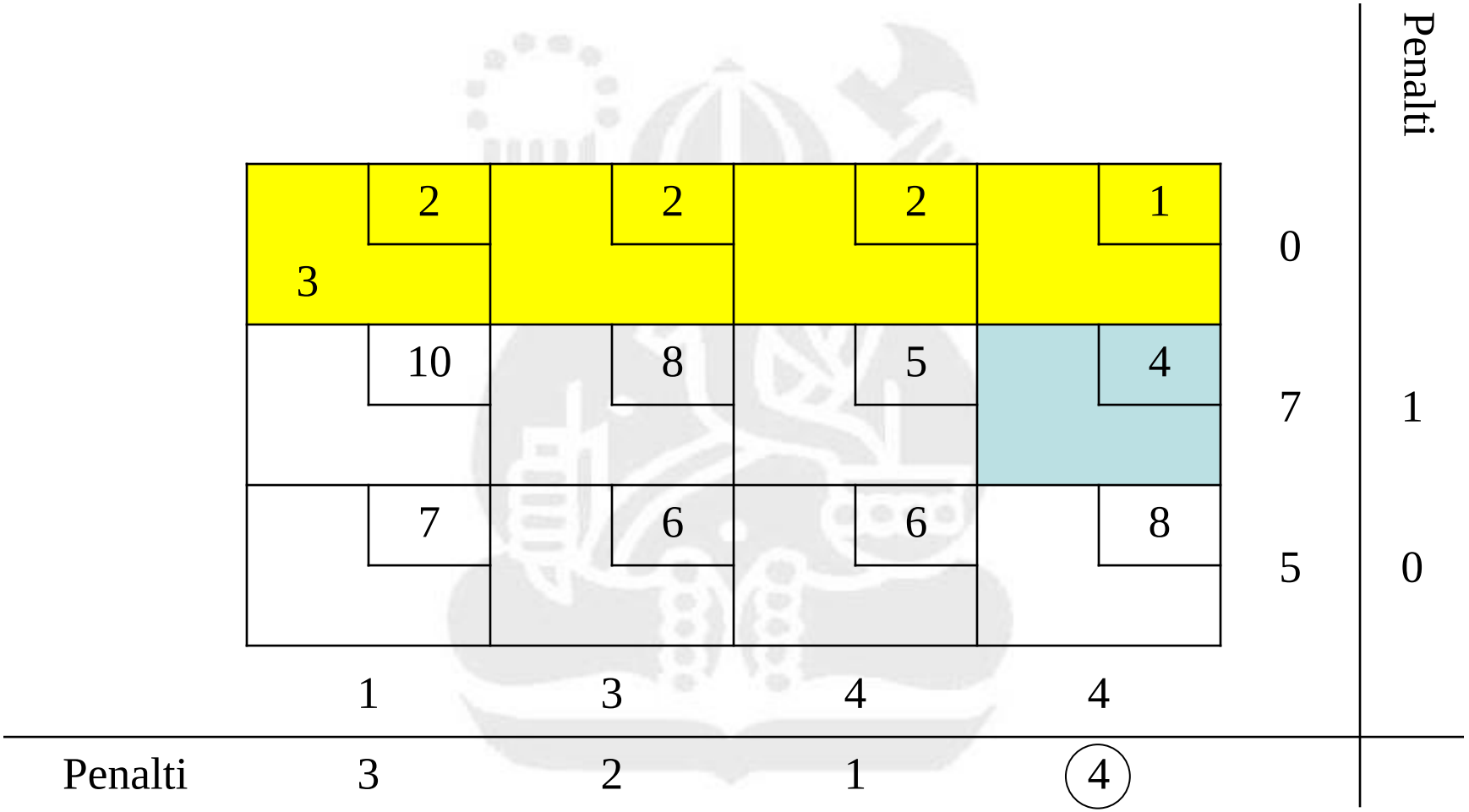


Biaya transportasi total $Z = 79$

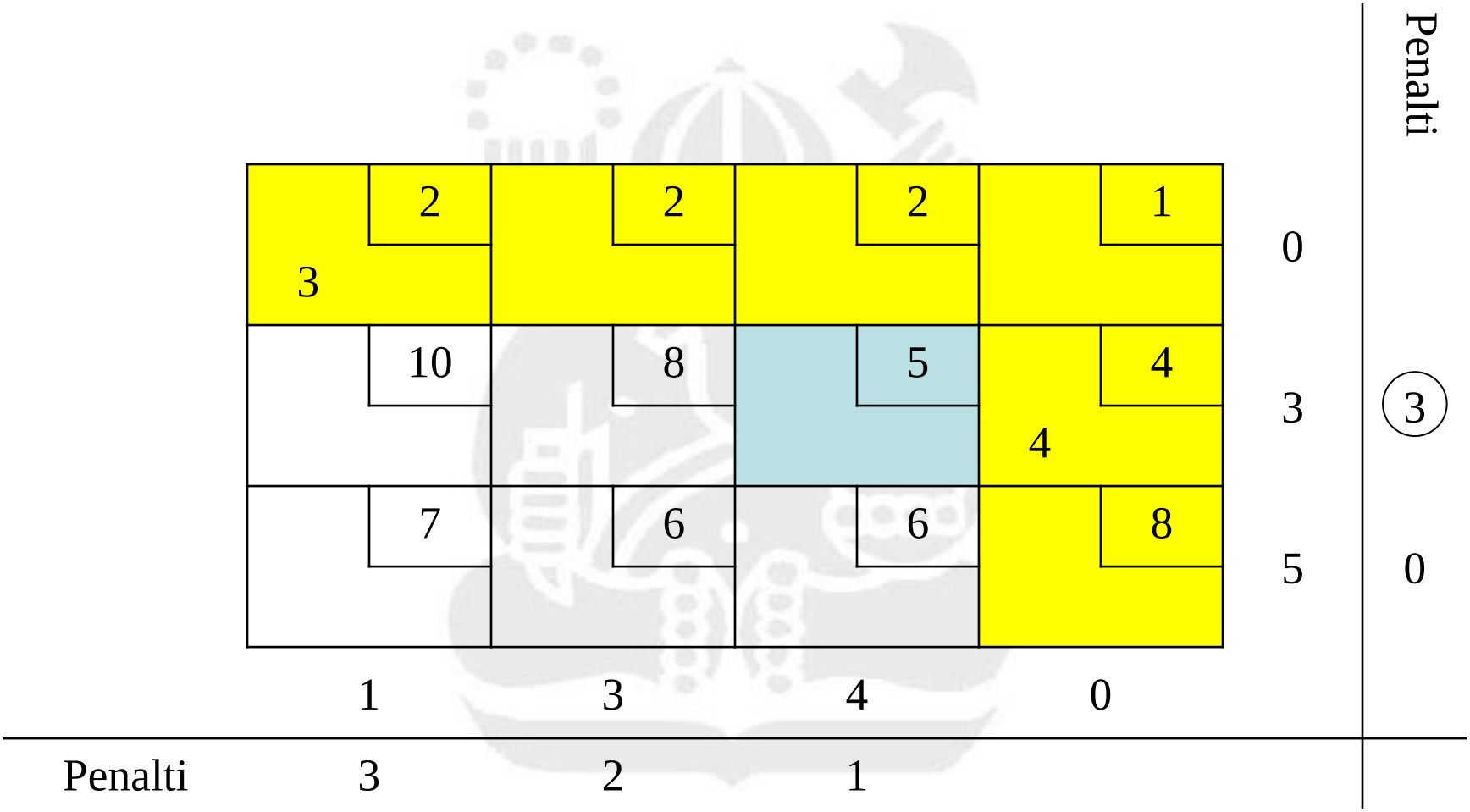
Vogel's Approximation Method (VAM) (1)



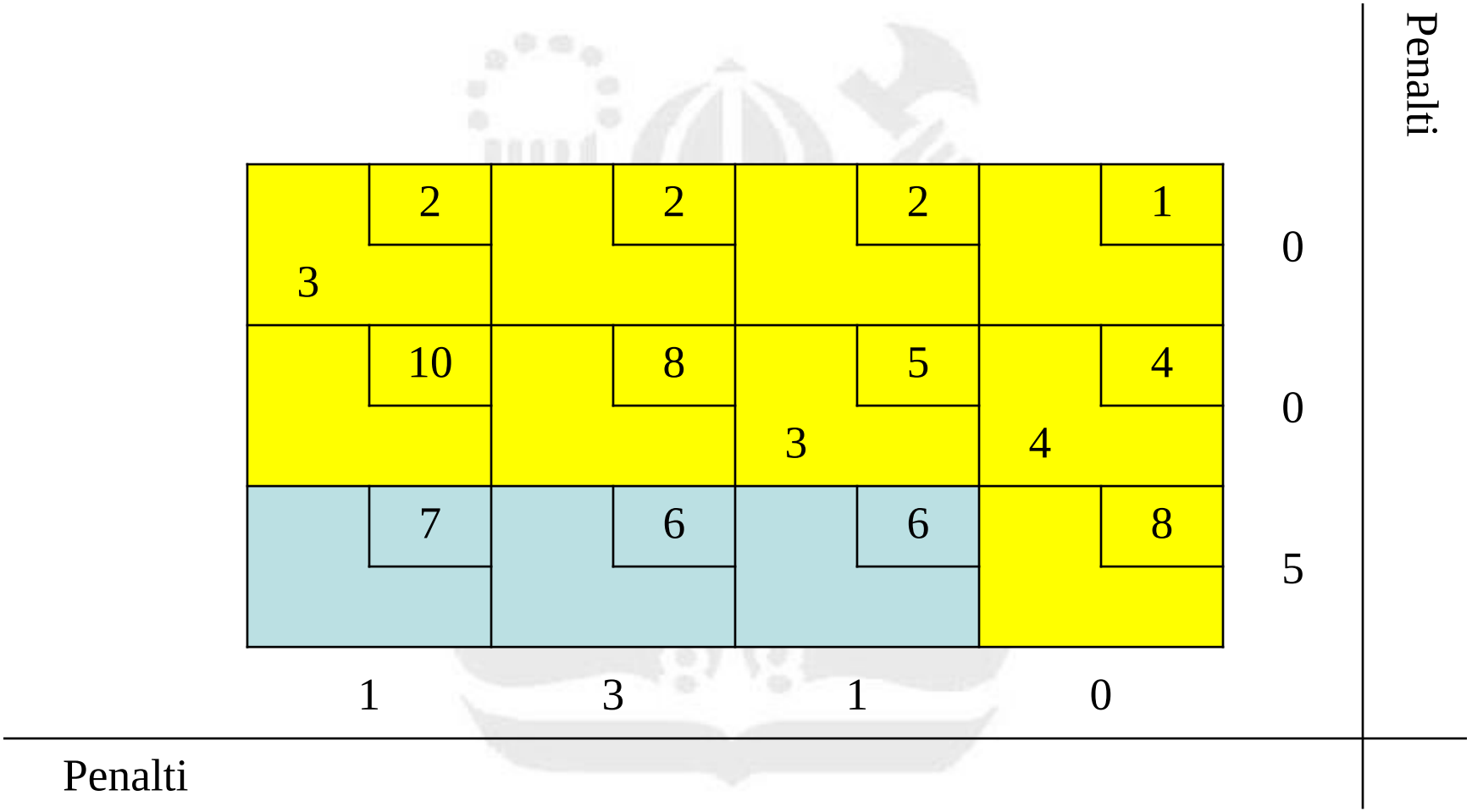
Vogel's Approximation Method (VAM) (2)



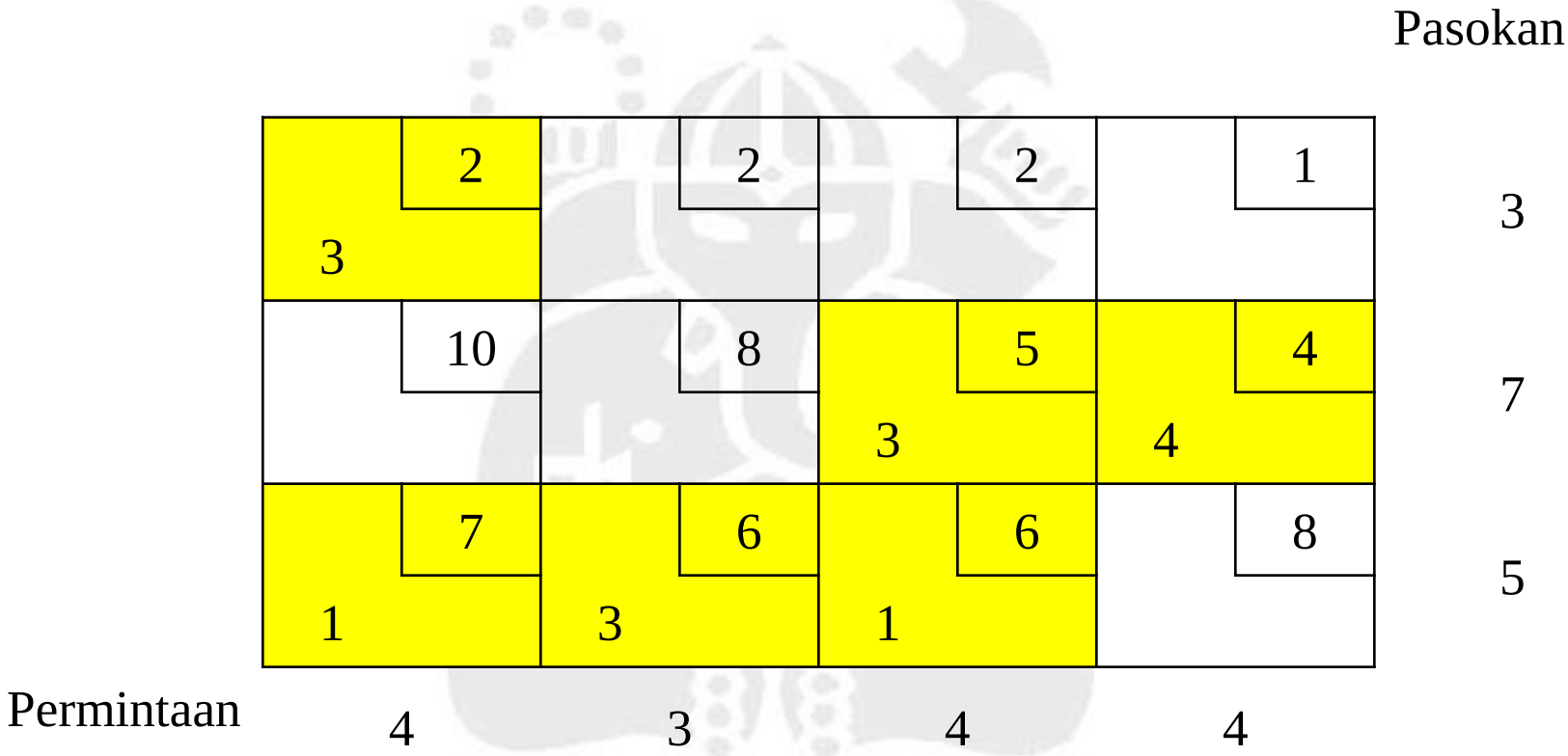
Vogel's Approximation Method (VAM) (3)



Vogel's Approximation Method (VAM) (4)



Vogel's Approximation Method (VAM) (5)
Solusi Basis Layak Awal



Biaya transportasi total $Z = 68$

Russell's Approximation Method (RAM) (1)

					$\bar{u}_i$
	2	2	2	1	2
	10	8	5	4	10
	7	6	6	8	8
$\bar{v}_i$	10	8	6	8	

$\Delta_{ij} = c_{ij} - \bar{u}_i - \bar{v}_j$

-10	-8	-6	-9	
-10	-10	-11	-14	
-11	-10	-8	-8	

				3	3
			4	7	3
				5	5
4	3	4	4		
4	3	4	0		

Russell's Approximation Method (RAM) (2)

					$\bar{u}_i$
	2	2	2		2
	10	8	5		10
	7	6	6		7
$\bar{v}_i$	10	8	6		

$$\Delta_{ij} = c_{ij} - \bar{u}_i - \bar{v}_j$$

-10	-8	-6		
-10	-10	-11		
-10	-9	-7		

				3	3
		3	4	7	0
				5	5
4	3	4	4		
4	3	1	0		

Russell's Approximation Method (RAM) (3)

$\bar{u}_i$

2	2	2		2
7	6	6		7
$\bar{v}_i$ 7	6	6		

$\Delta_{ij} = c_{ij} - \bar{u}_i - \bar{v}_j$

-7	-6	-6		
-7	-7	-7		

3				3	0
		3	4	7	0
				5	5
4	3	4	4		
1	3	1	0		

Russell's Approximation Method (RAM) (4)

$\bar{u}_i$

	7	6	6	7
$\bar{v}_i$	7	6	6	

$$\Delta_{ij} = c_{ij} - \bar{u}_i - \bar{v}_j$$

	-7	-7	-7	

	3				3 0
			3	4	7 0
	1				5 4
	4	3	4	4	
	0	3	1	0	

Russell's Approximation Method (RAM) (5)

$\bar{u}_i$

	6	6		6
$\bar{v}_i$	6	6		

$$\Delta_{ij} = c_{ij} - \bar{u}_i - \bar{v}_j$$

	-6	-6		

3				3	0
		3	4	7	0
1	3			5	1
4	3	4	4		
0	0	1	0		

Russell's Approximation Method (RAM) (6)

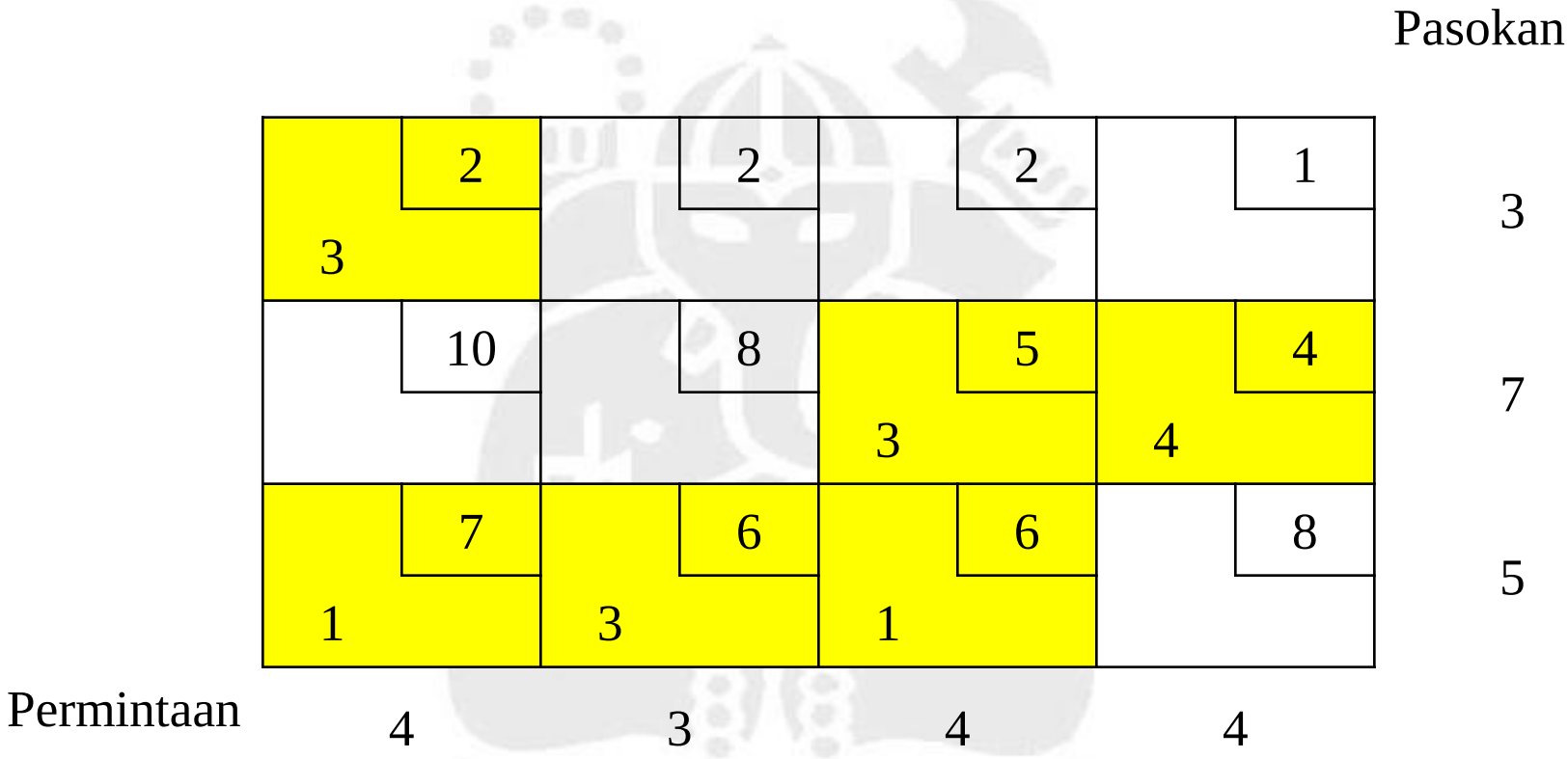
$\bar{u}_i$

			6	6
$\bar{v}_i$			6	

$\Delta_{ij} = c_{ij} - \bar{u}_i - \bar{v}_j$

			-6		
3				3	0
		3	4	7	0
1	3	1		5	0
4	3	4	4		
0	0	0	0		

Russell's Approximation Method (RAM) (7)

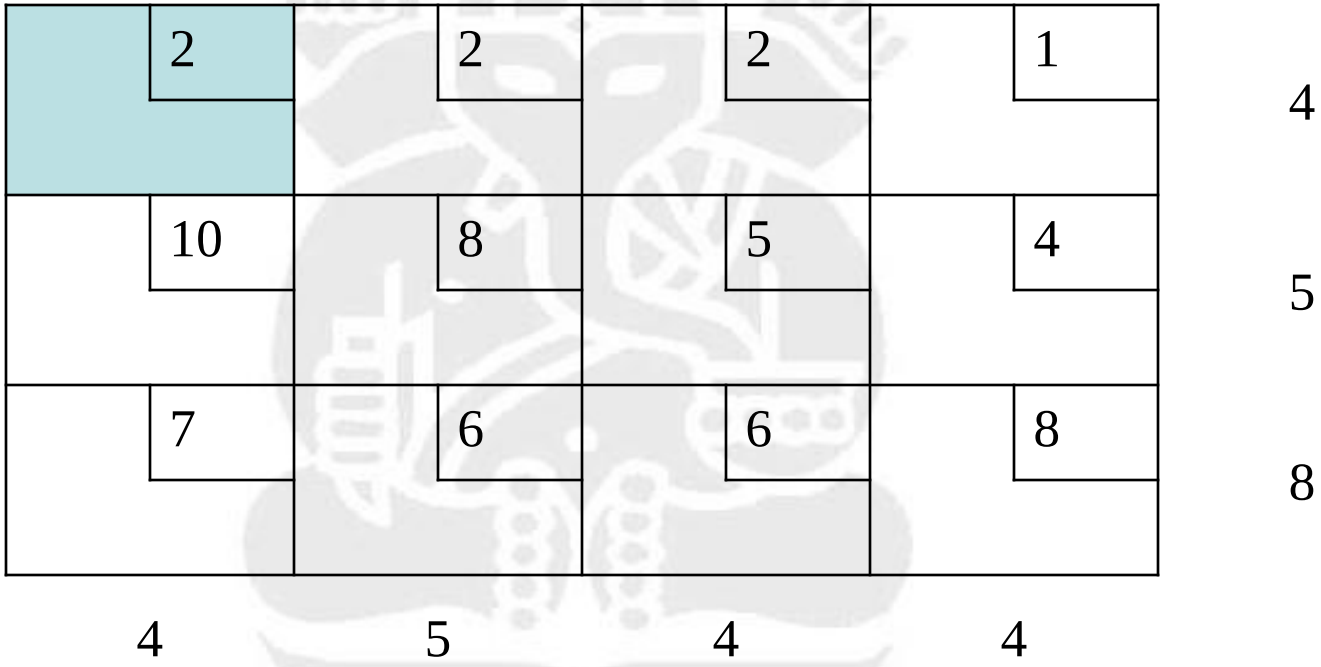


Biaya transportasi total $Z = 68$

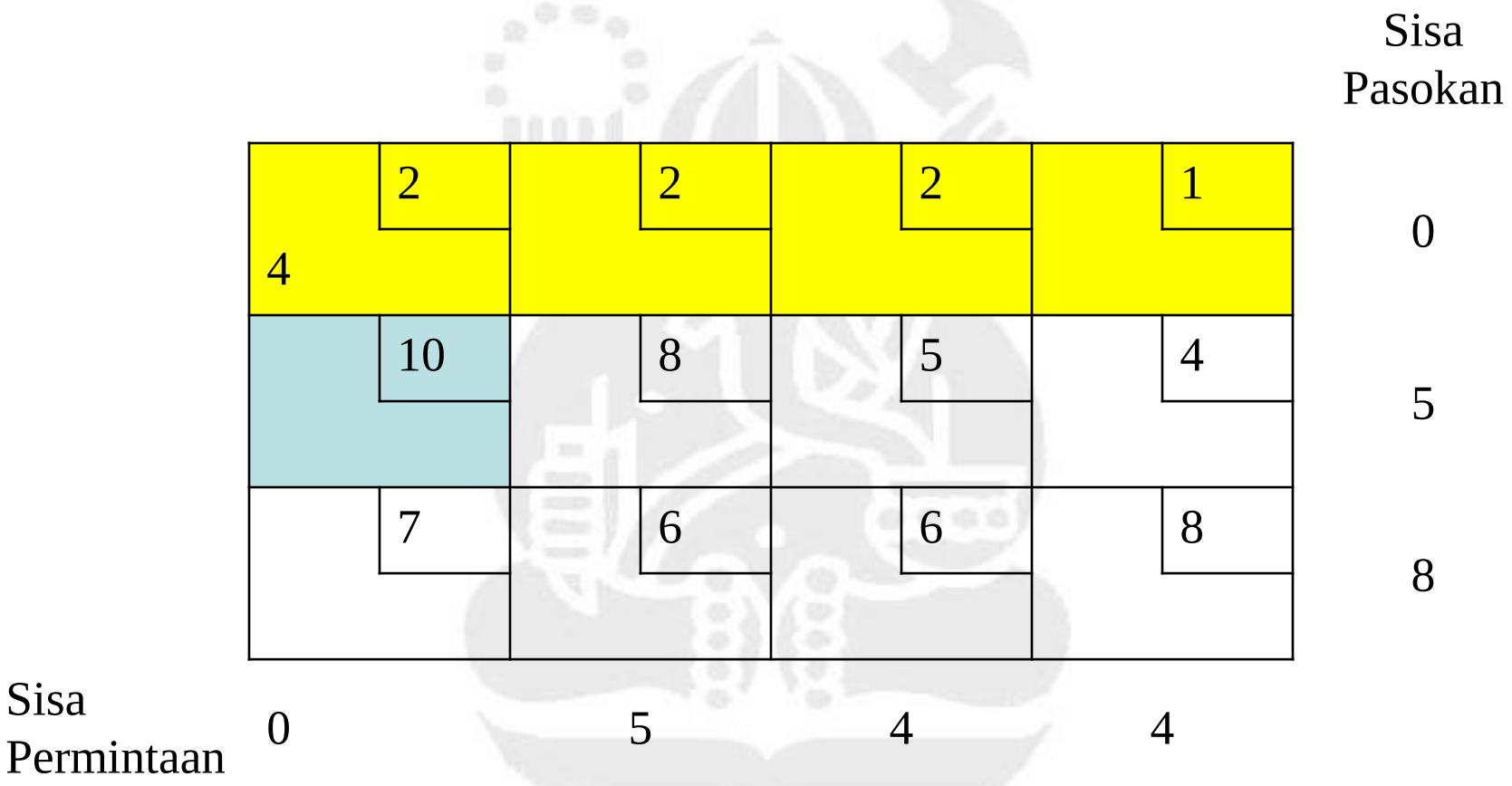
Degenerasi

- Solusi basis layak dari masalah transportasi dikatakan degenerasi (*degenerate*) jika satu atau lebih variabel basis mempunyai nilai nol.
- Solusi basis dapat menjadi degenerasi jika sisa pasokan dan sisa permintaan adalah sama untuk variabel yang akan dipilih menjadi basis.
- Jumlah solusi basis dalam masalah transportasi harus : $m + n - 1$ (m = jumlah baris, n = jumlah kolom)

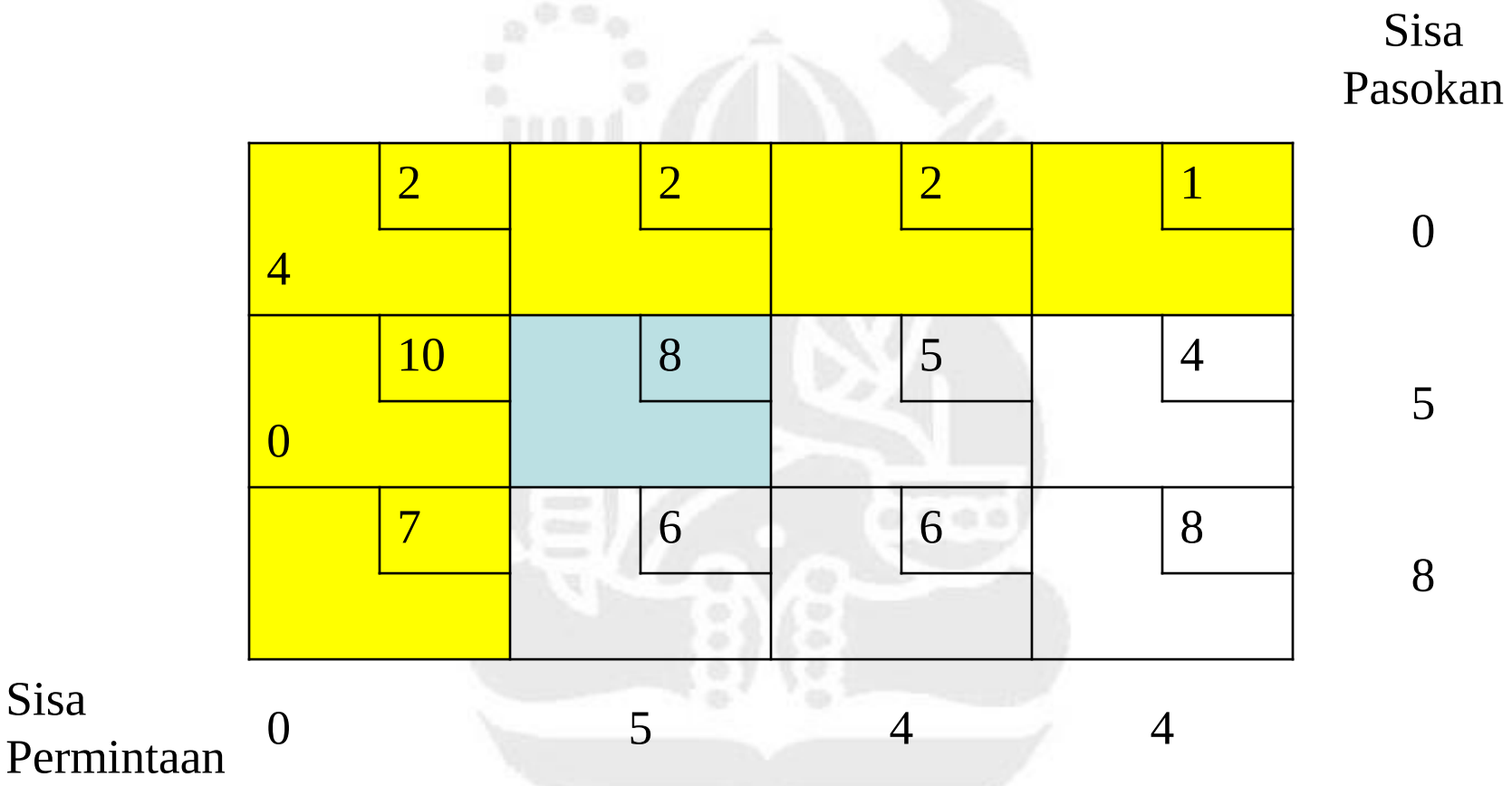
Ilustrasi Degenerasi (1)



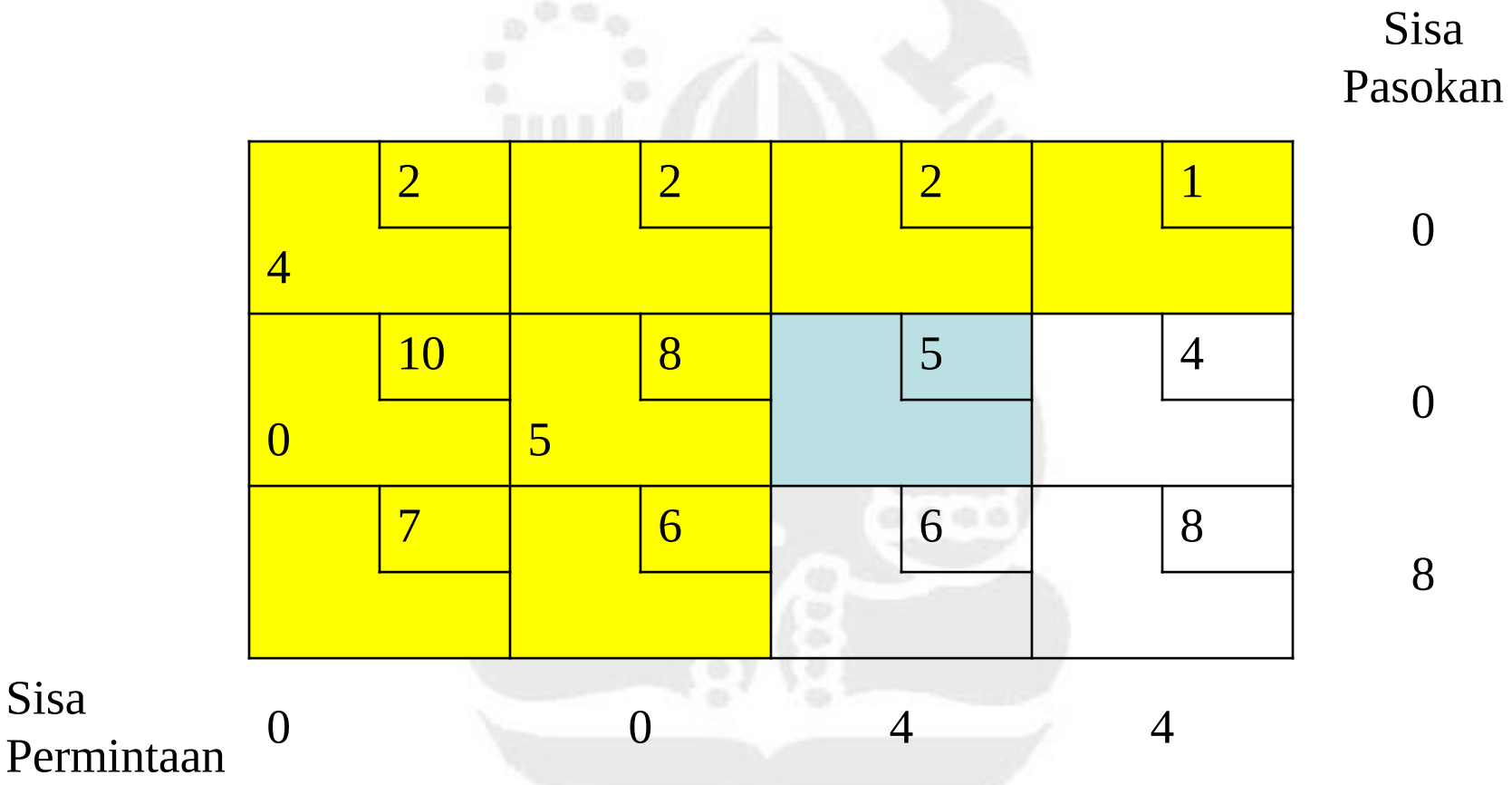
Ilustrasi Degenerasi (2)



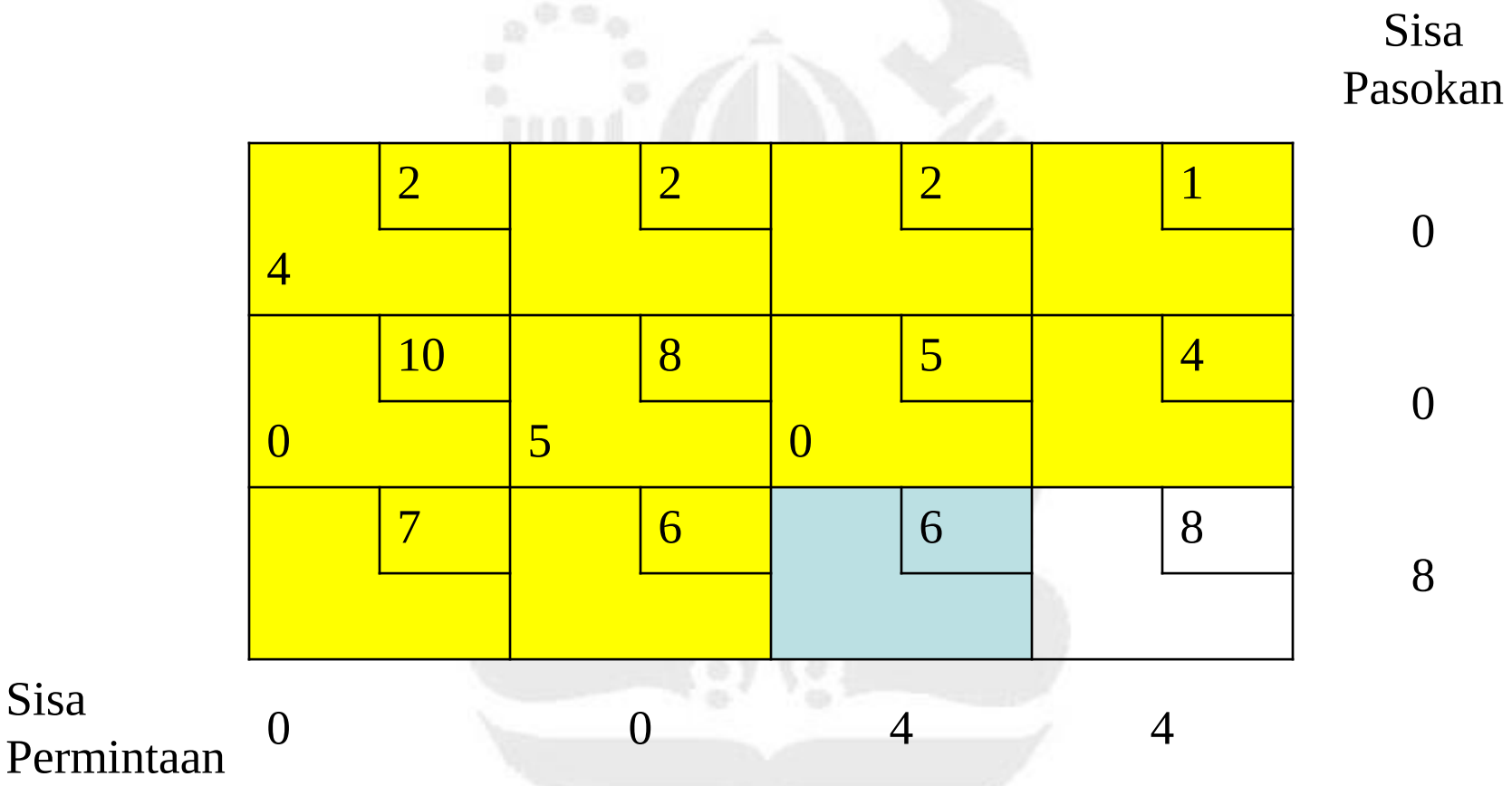
Ilustrasi Degenerasi (3)



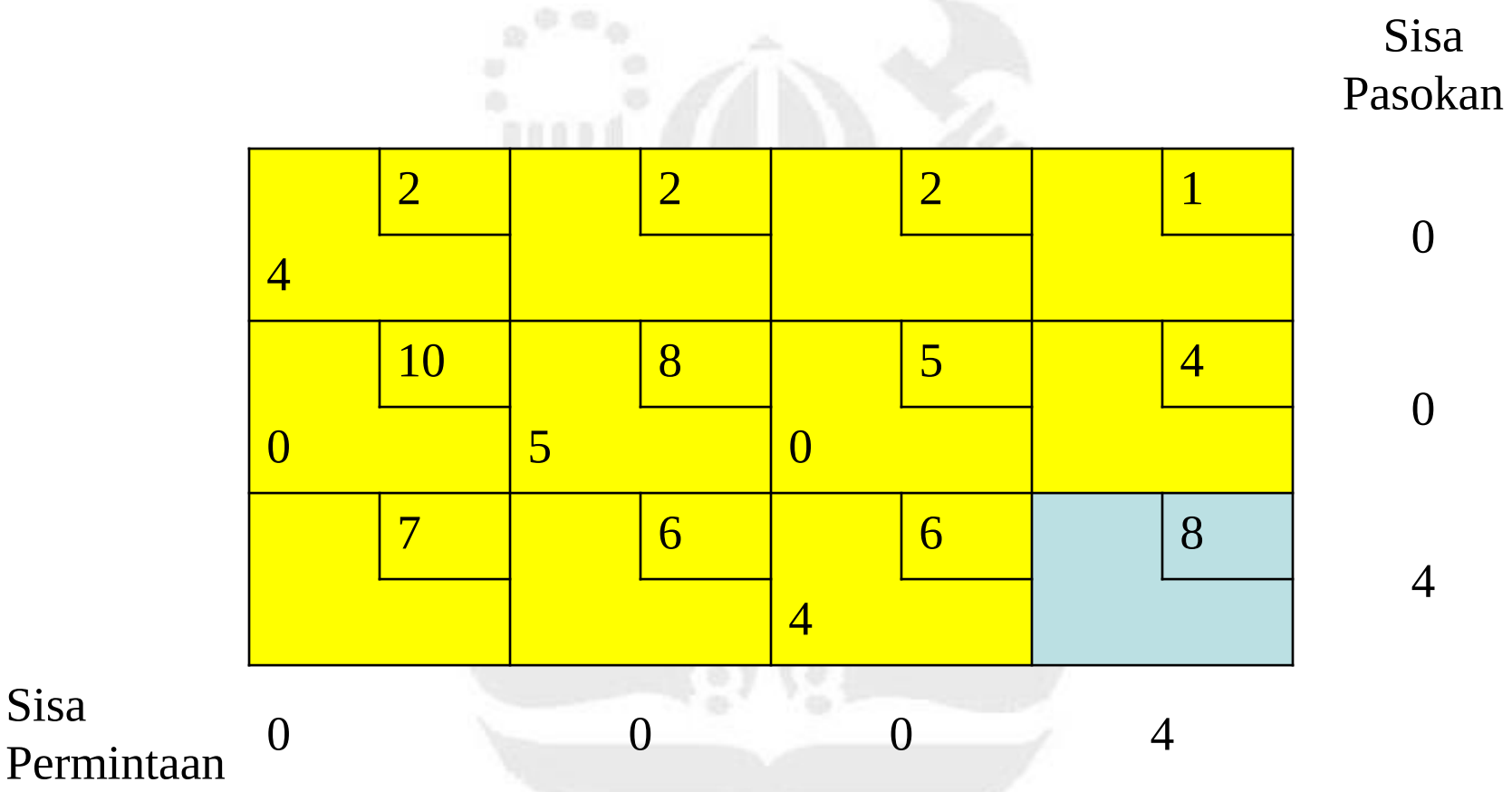
Ilustrasi Degenerasi (4)



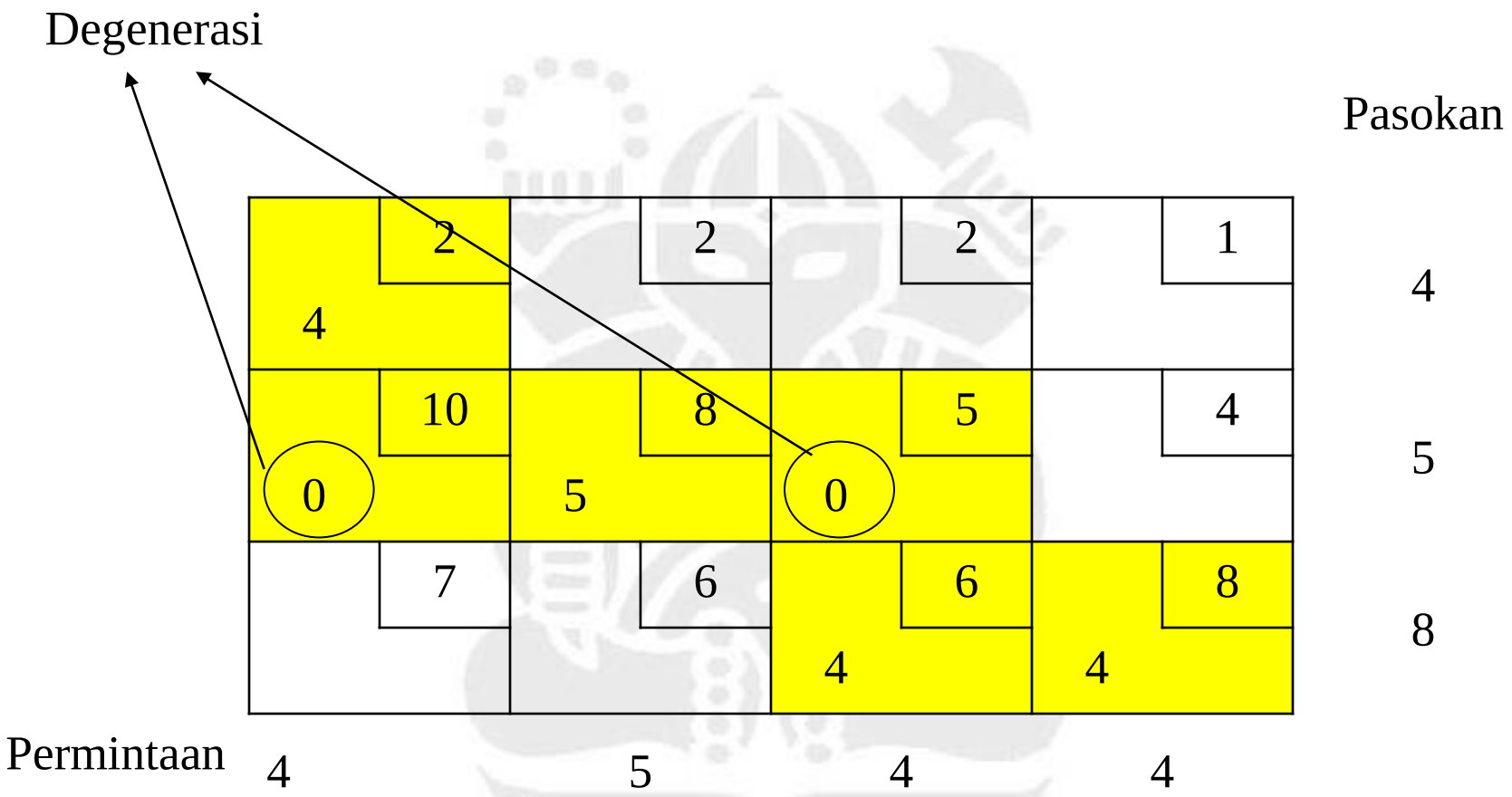
Ilustrasi Degenerasi (5)



Ilustrasi Degenerasi (6)



Ilustrasi Degenerasi (7)



Perbaikan Solusi Basis Layak Awal

- Perbaikan solusi basis layak awal
  - Pemeriksaan optimalitas
  - Penentuan solusi basis layak yang baru
- Metoda:
  - Metoda *u-v* atau MODI (*Modified Distribution Method)*

Metode *u*-*v* (1)

Untuk sebarang solusi basis layak, tentukan nilai u_i (untuk semua i) dan v_j (untuk semua j) sedemikian hingga

$$u_i + v_j = c_{ij} \qquad \text{untuk setiap variabel basis } x_{ij}$$

(Nilai u_i dan v_j bisa positif, negatif atau nol).

Untuk variabel non basis:

$$\bar{c}_{ij} = c_{ij} - \left(u_i + v_j\right)$$

Metode *u*-*v* (2)

Untuk variabel non basis:

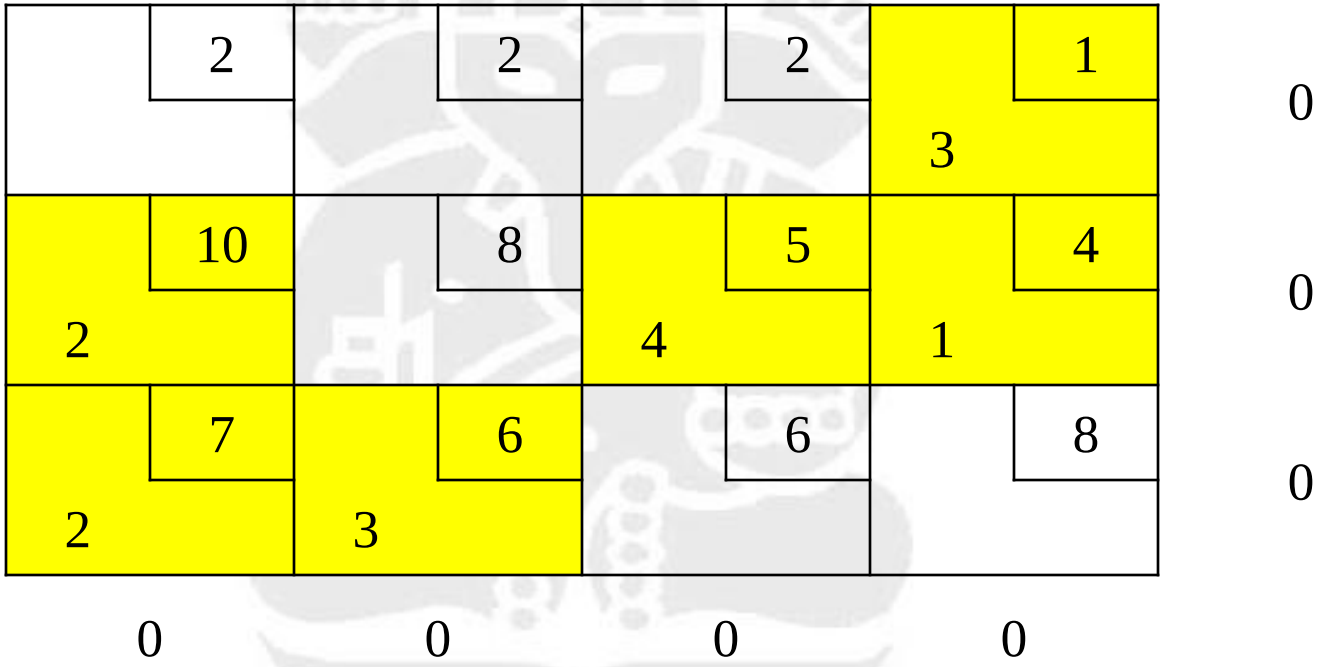
$$\bar{c}_{ij} = c_{ij} - \left(u_i + v_j\right)$$

Kondisi optimalitas (masalah minimasi) terjadi apabila

$$\bar{c}_{ij} = c_{ij} - \left(u_i + v_j\right) \geq 0$$ untuk semua variabel non basis

Jika kondisi belum optimal, variabel yang masuk basis adalah yang mempunyai $\bar{c}_{ij}$ paling negatif (masalah minimasi)

Misal Diberikan Solusi Basis Layak Awal dengan *Least Cost Method*



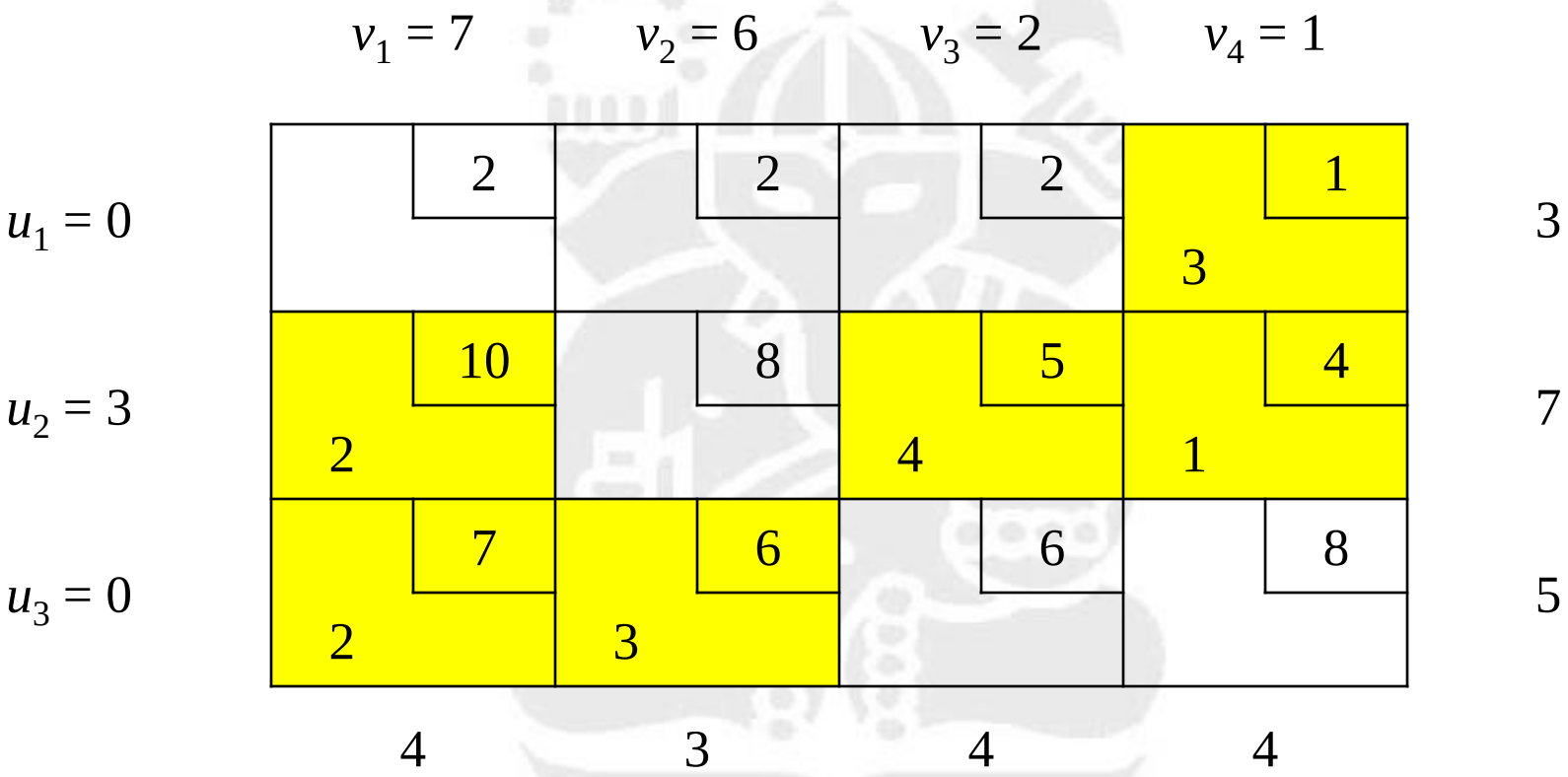
Penerapan Metode *u-v*

$$u_1 + v_4 = 1$$
$$u_2 + v_1 = 10$$
$$u_2 + v_3 = 5$$
$$u_2 + v_4 = 4$$
$$u_3 + v_1 = 7$$
$$u_3 + v_2 = 6$$

Enam persamaan dengan tujuh variabel yang tak diketahui → terdapat tak hingga solusi yang mungkin

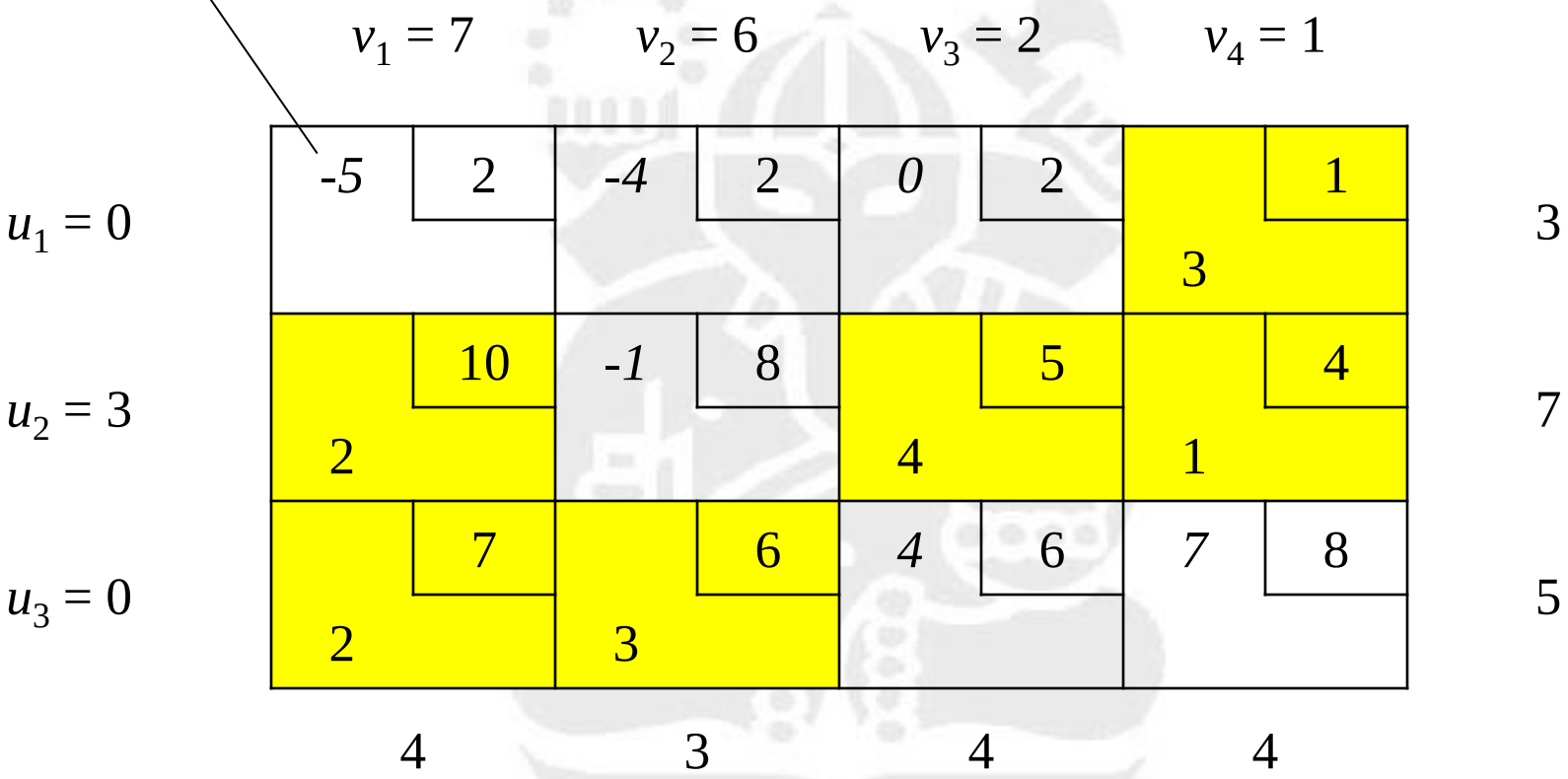
Untuk mendapatkan solusi, suatu nilai variabel tertentu dapat ditetapkan sebarang, dan nilai yang lain dapat dipecahkan. Misalnya, $u_1 = 0$

Pemeriksaan Optimalitas (1)



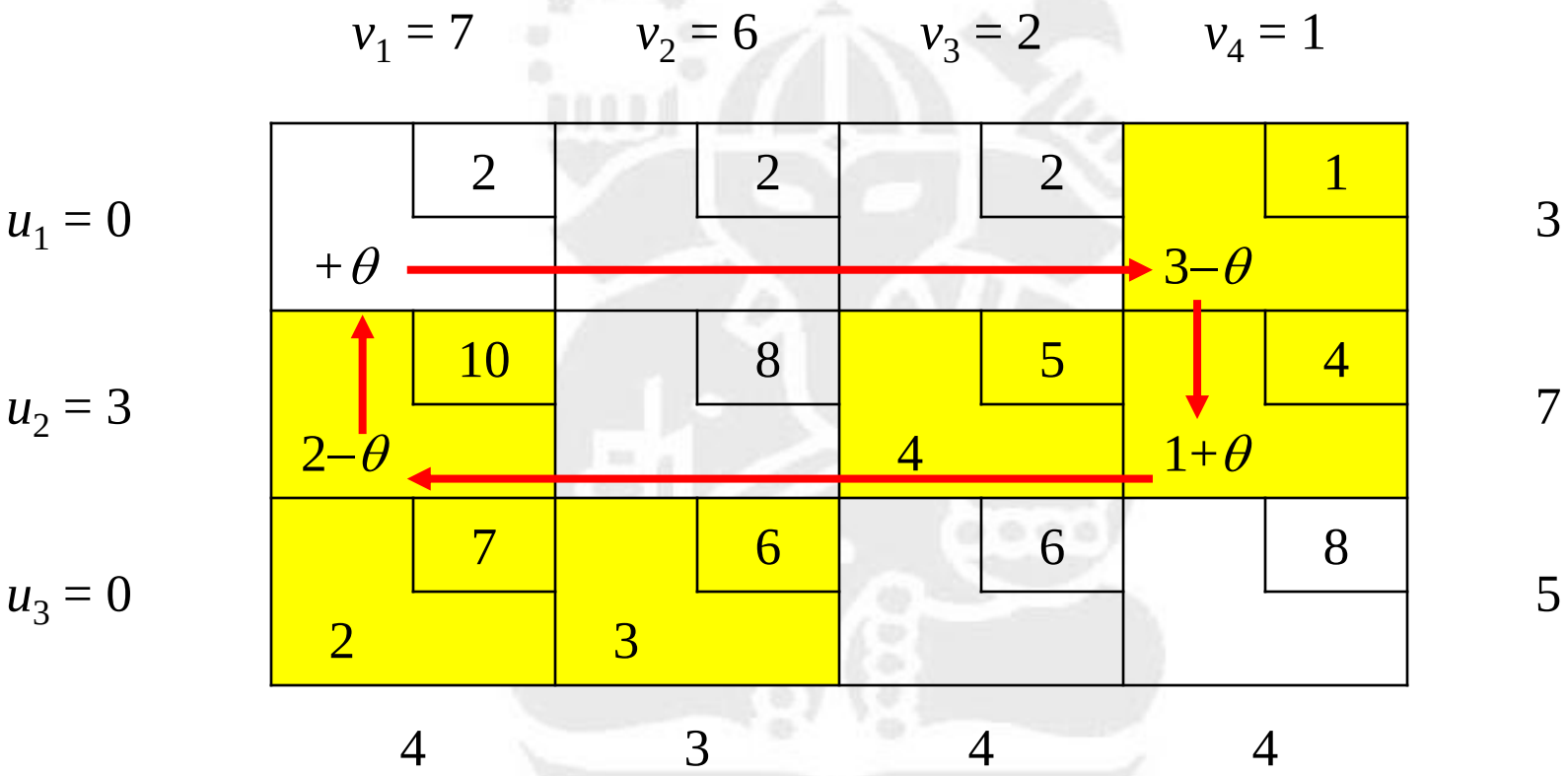
Pemeriksaan Optimalitas (2)

$$\bar{c}_{ij} = c_{ij} - (u_i + v_j)$$



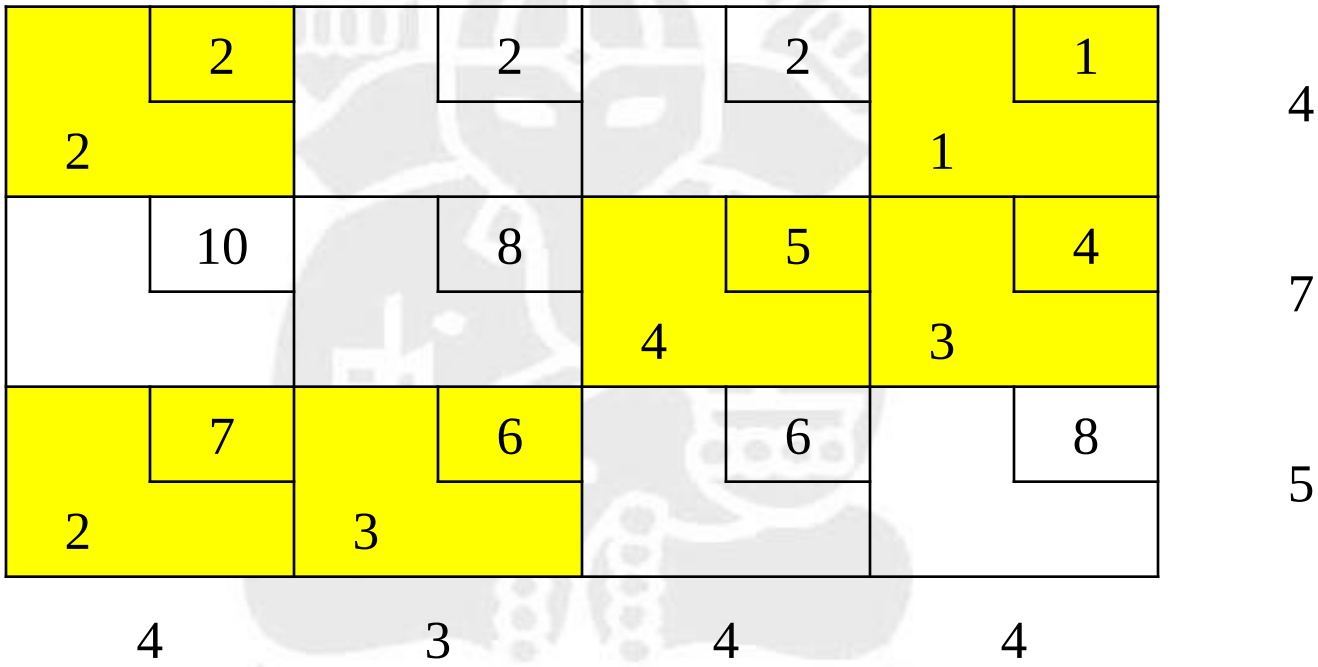
x_{11} masuk basis

Penentuan variabel keluar basis



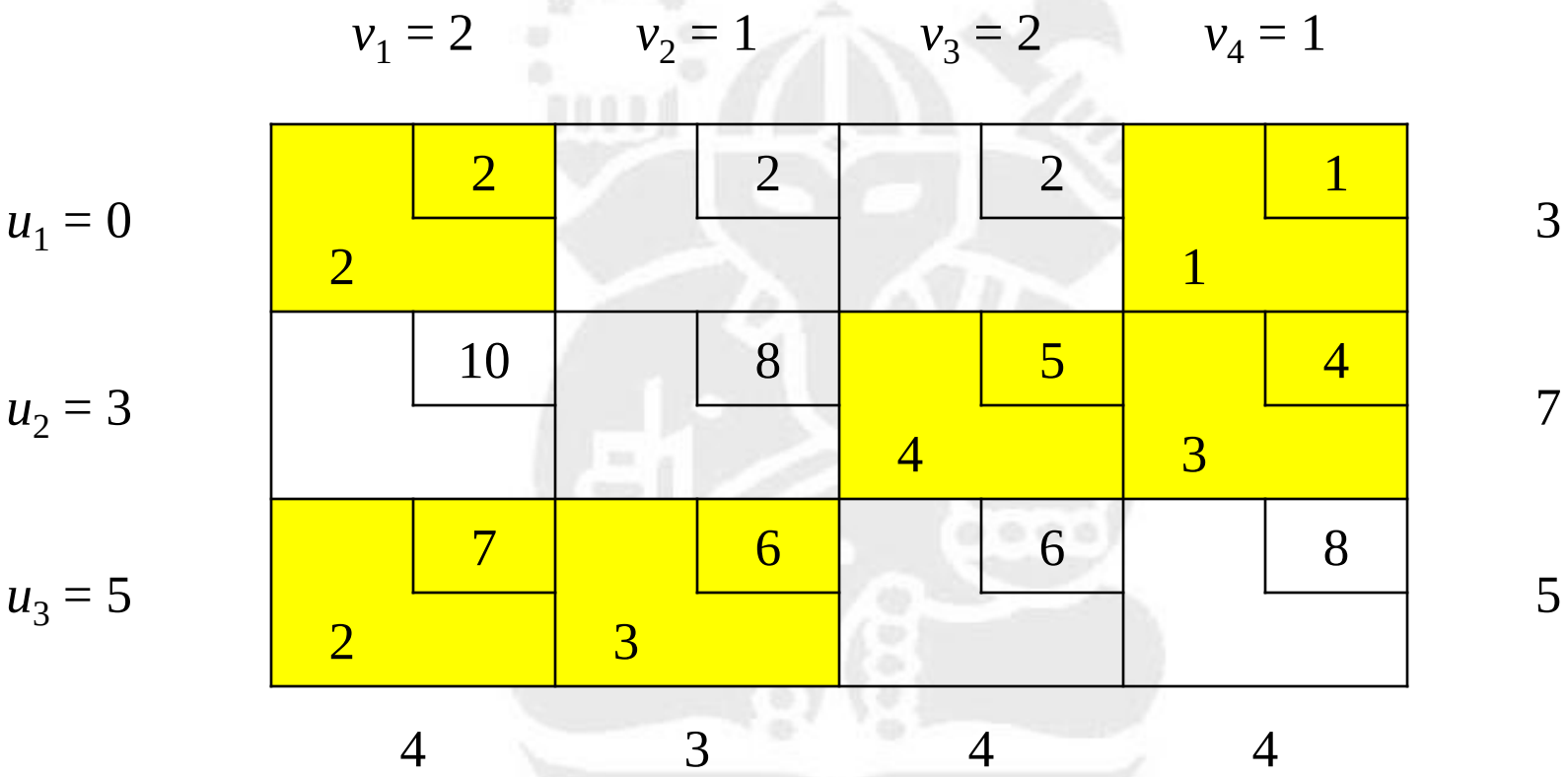
$\theta = \min(3, 2) = 2 \rightarrow x_{21}$ keluar basis

Solusi Iterasi ke-1

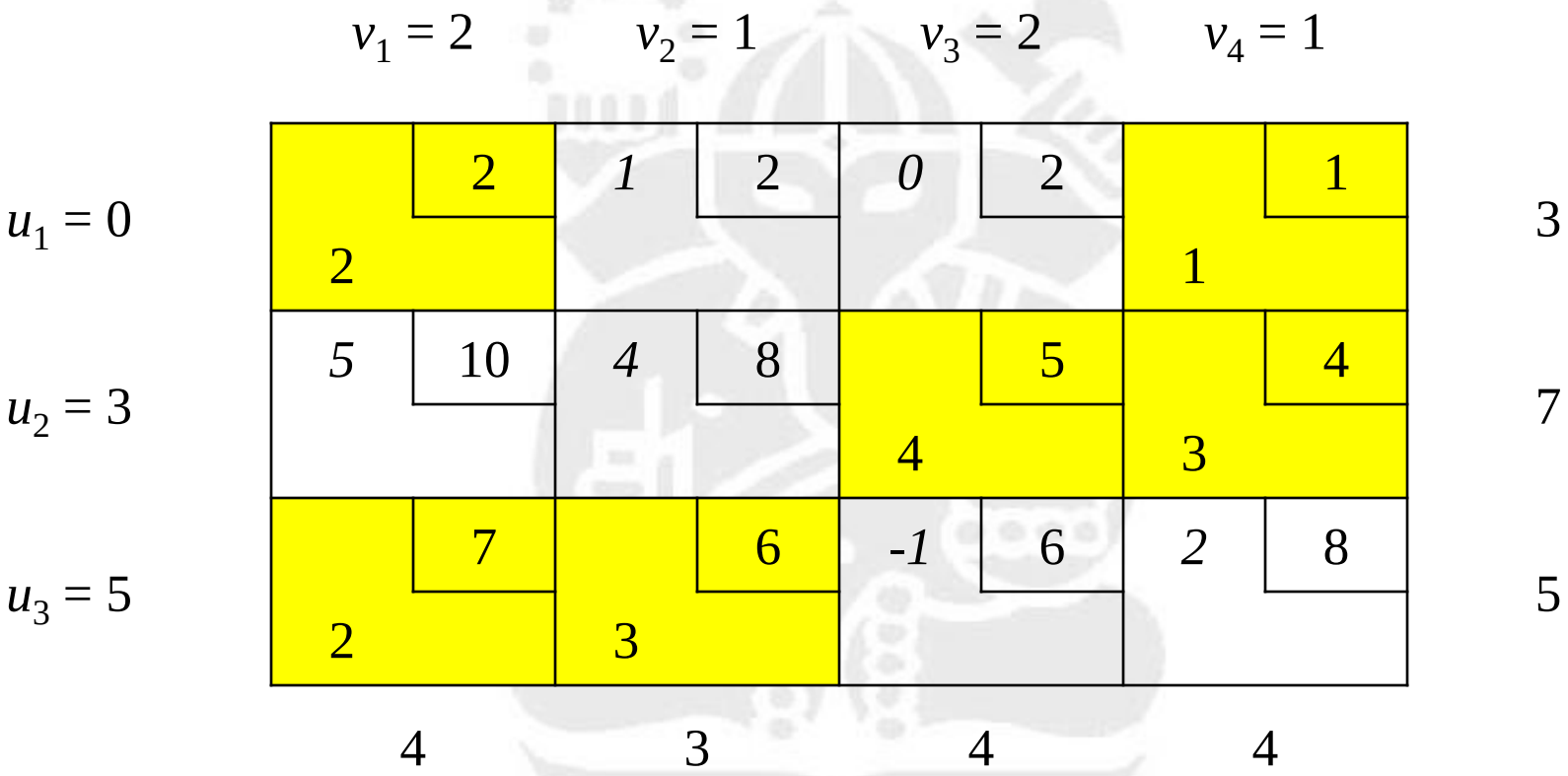


Biaya transportasi total $Z = 69$

Pemeriksaan Optimalitas (1)

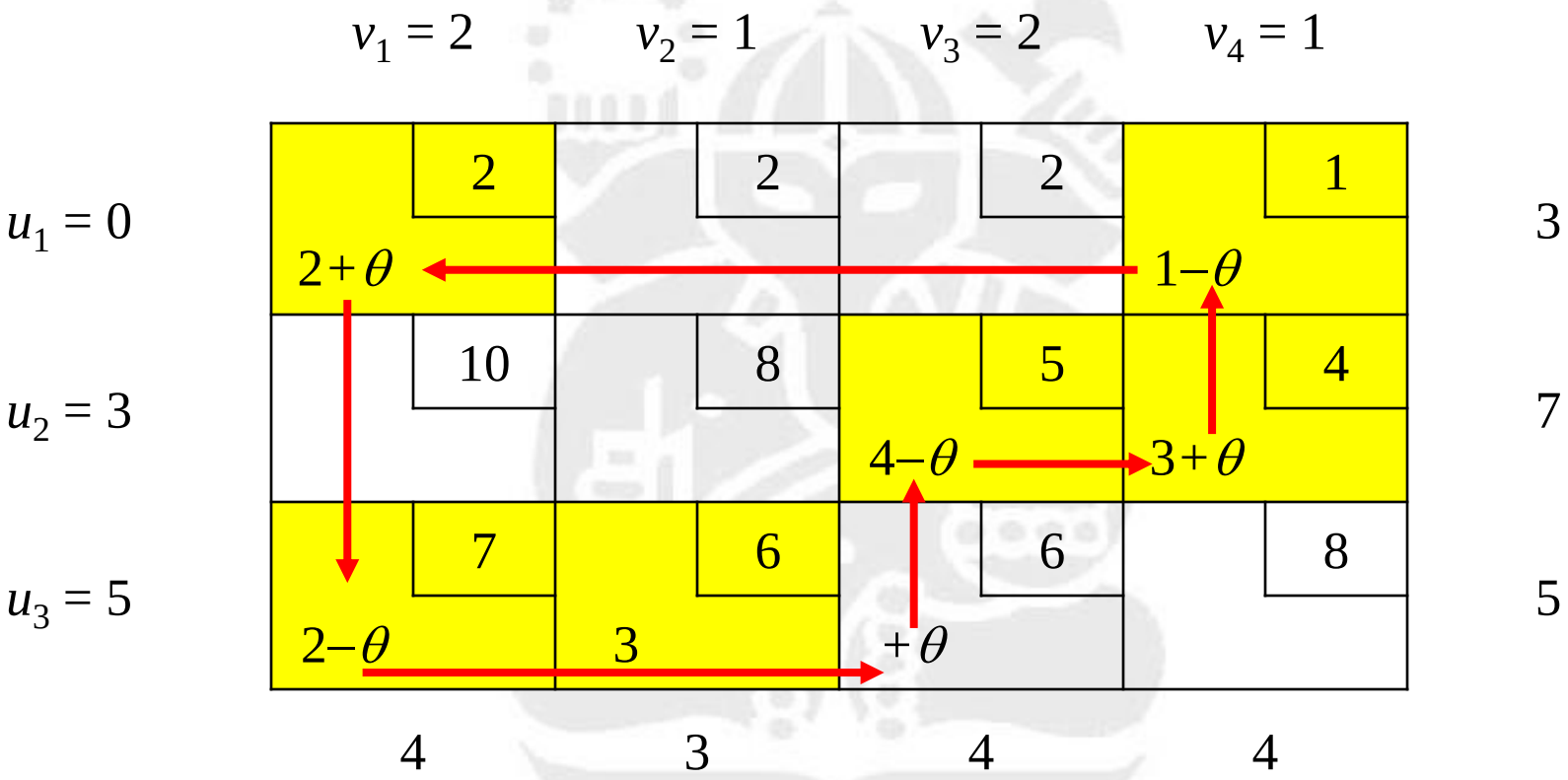


Pemeriksaan Optimalitas (2)



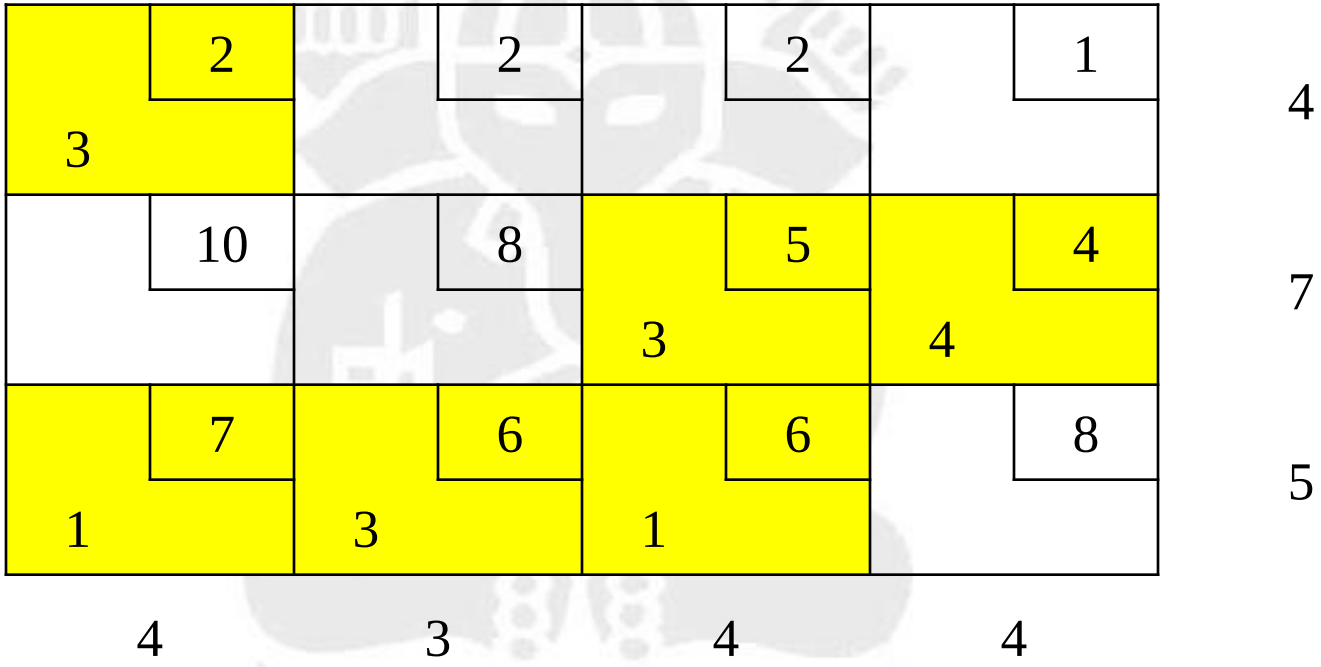
x_{33} masuk basis

Penentuan variabel keluar basis



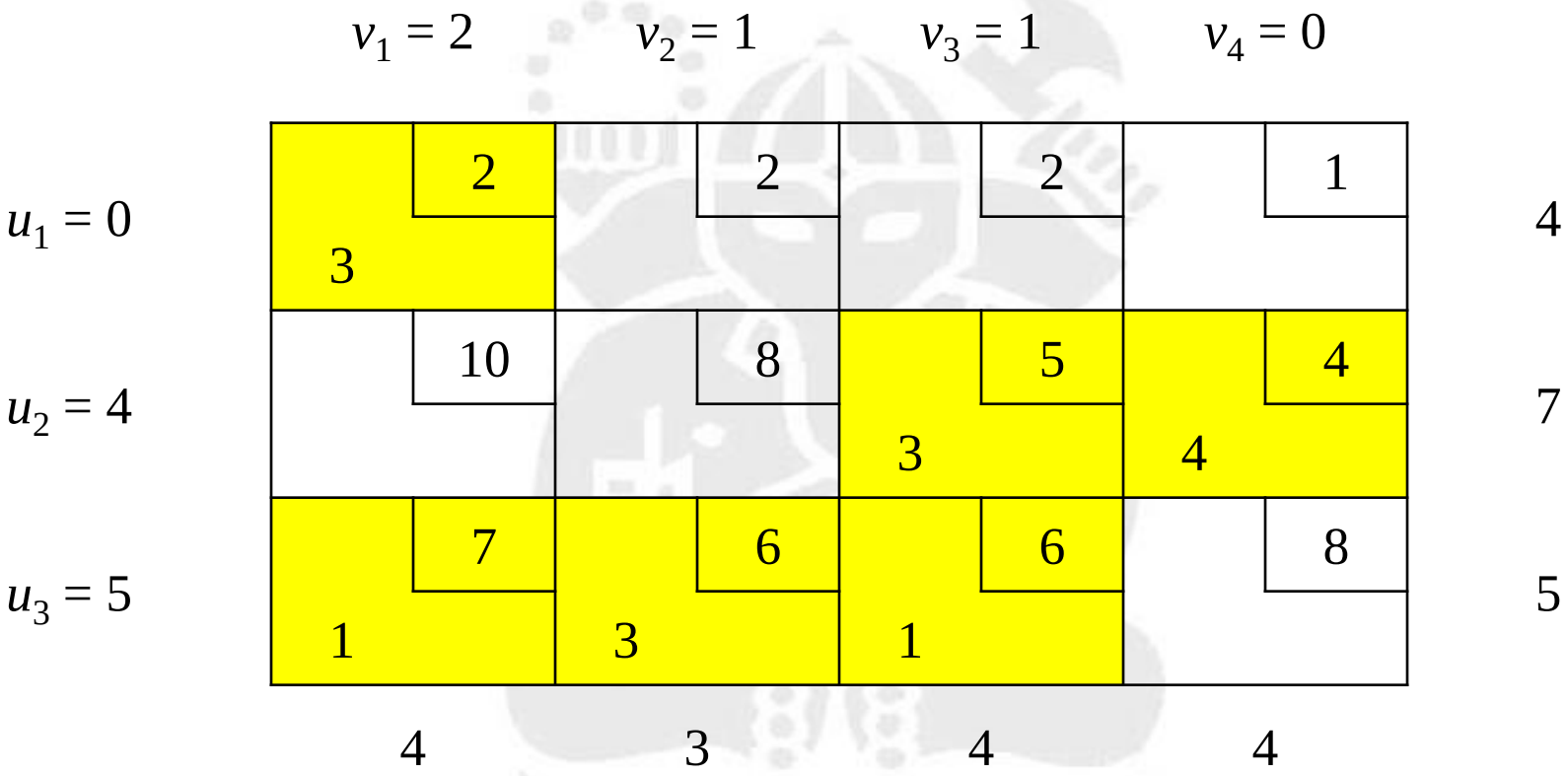
$\theta = \min(1, 4, 2) = 1 \rightarrow x_{14}$ keluar basis

Solusi Iterasi Ke-2



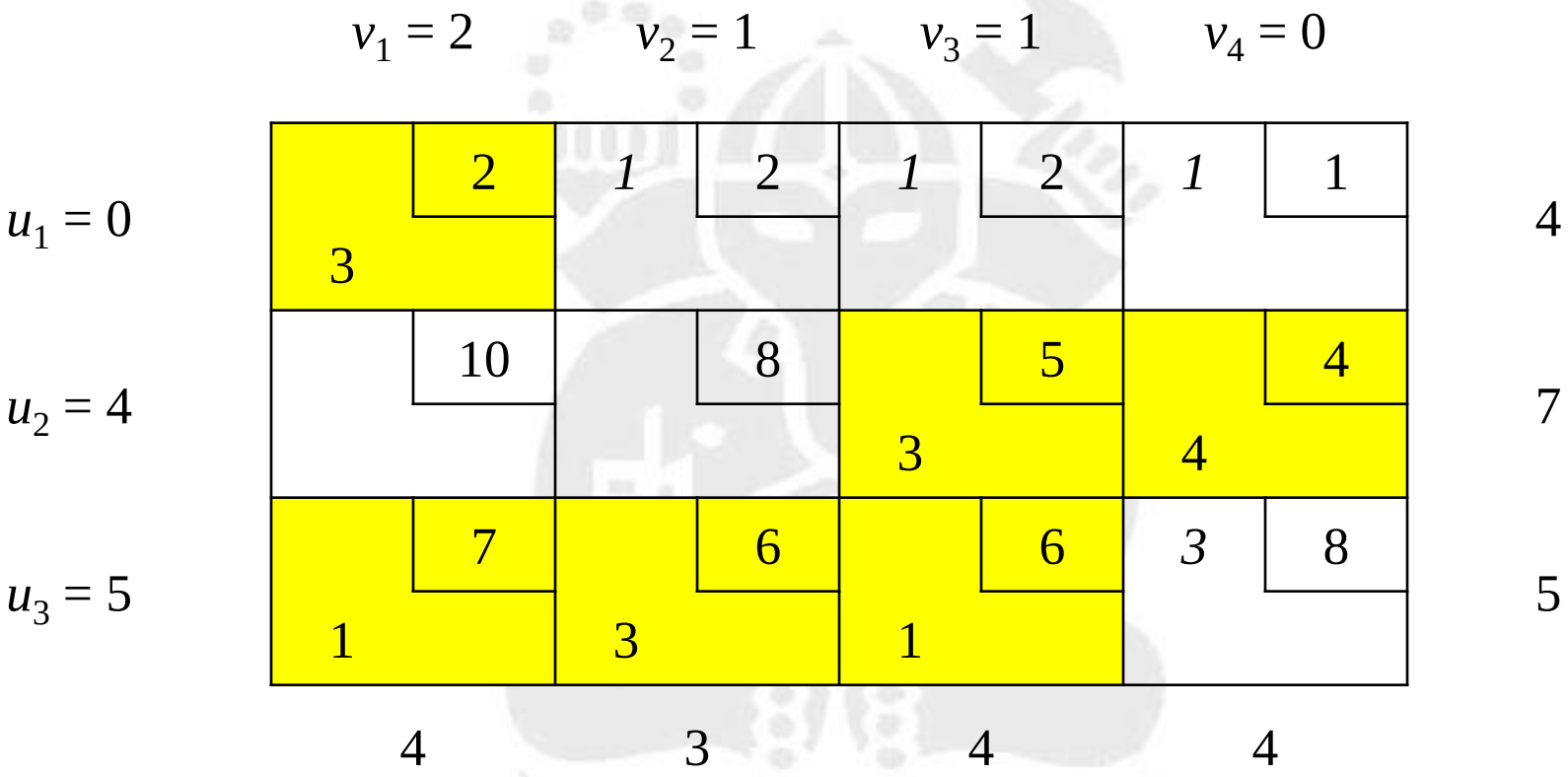
Biaya transportasi total $Z = 68$

Pemeriksaan Optimalitas (1)



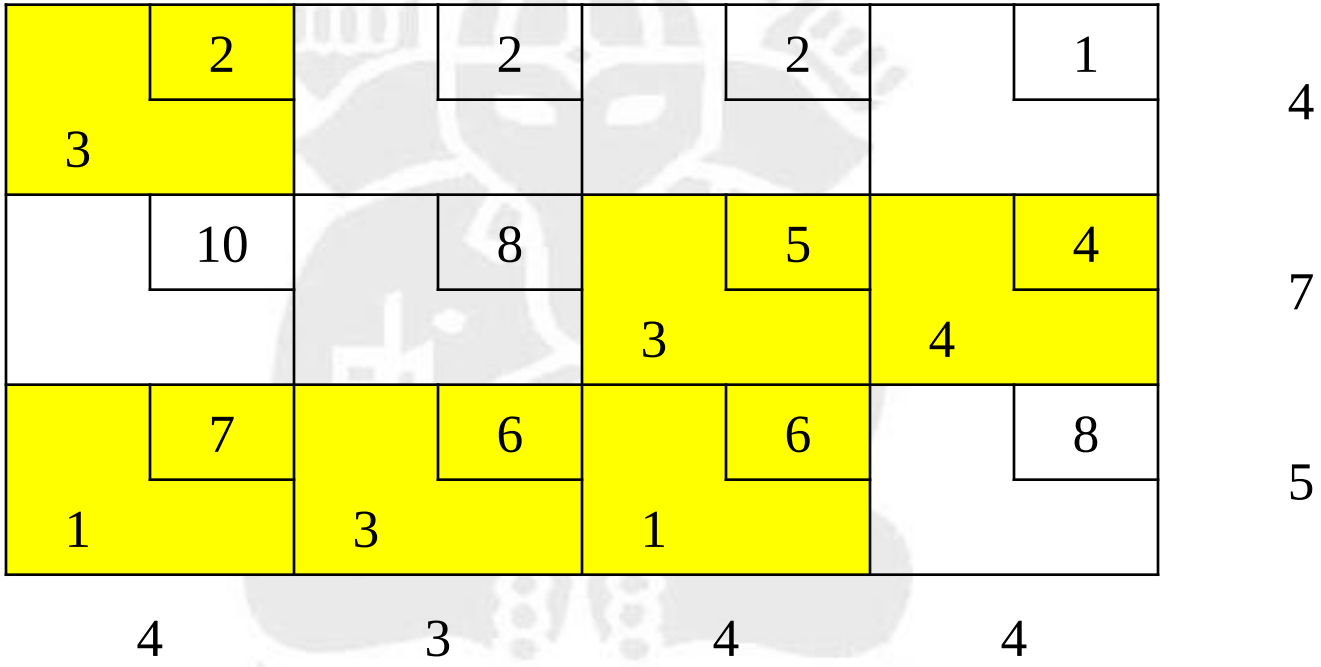
Biaya transportasi total $Z = 68$

Pemeriksaan Optimalitas (2)



Solusi optimal

Solusi Optimal



Biaya transportasi total $Z = 68$

Masalah Maksimisasi

Kondisi optimal :

→ Koefisien fungsi tujuan relatif untuk variabel non basis tak positif

$$\bar{c}_{ij} = c_{ij} - \left(u_i + v_j\right) \leq 0$$

Penentuan variabel non basis yang masuk basis

→Pilih variabel non basis dengan koefisien fungsi tujuan relatif paling positif